The STAC Field Guide

An adventure of your own making

FOSS4G | Auckland, NZ | Nov 2025

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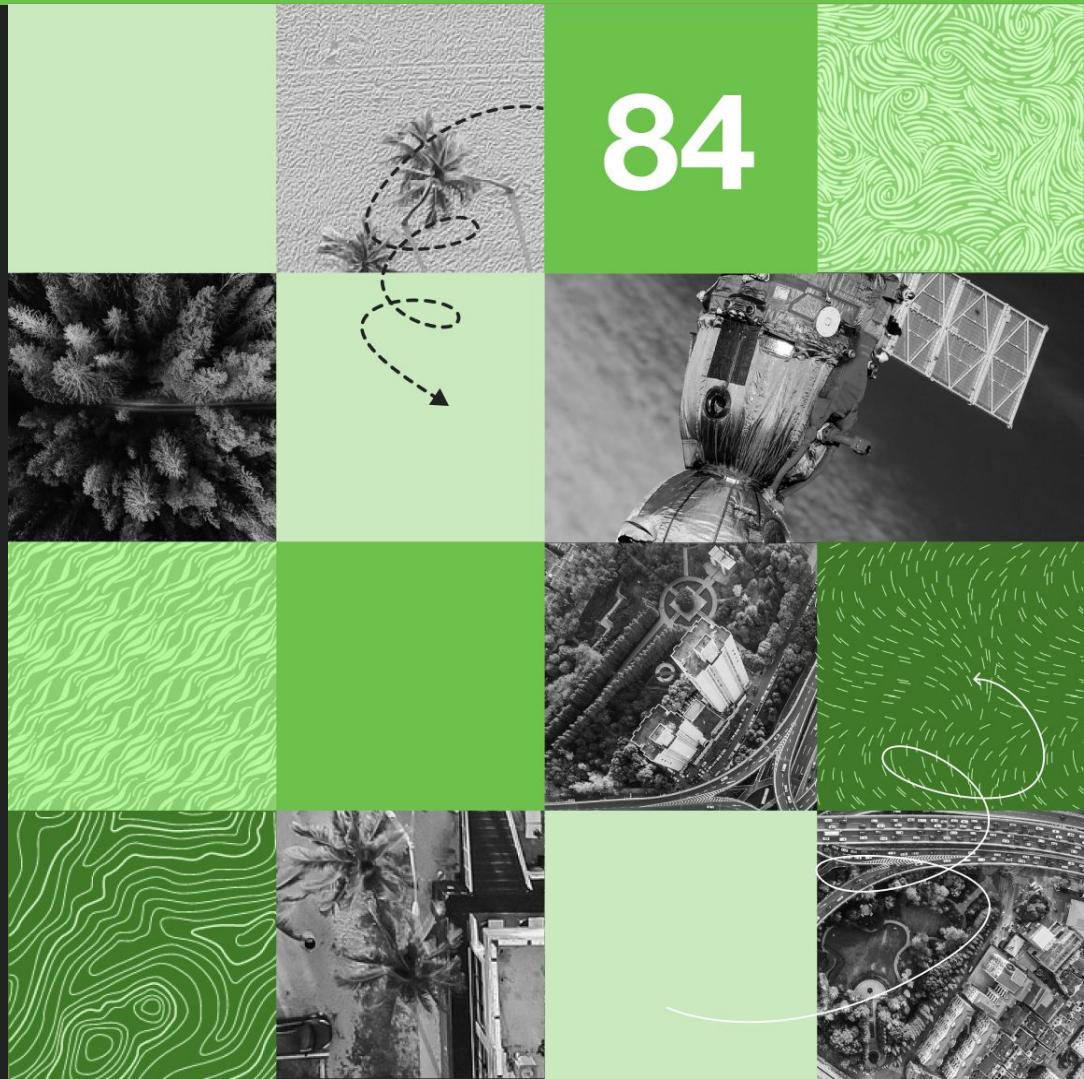
Element 84



STAC 101: What You Really Need to Know

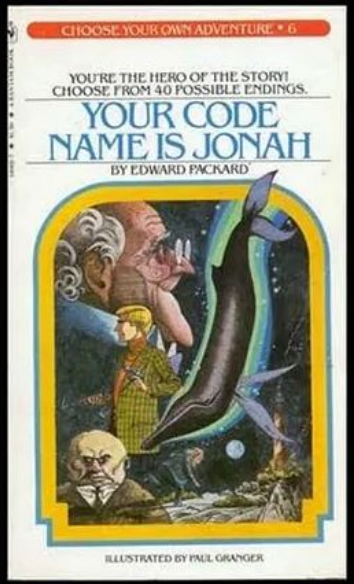
1:30 PM 25min WG403

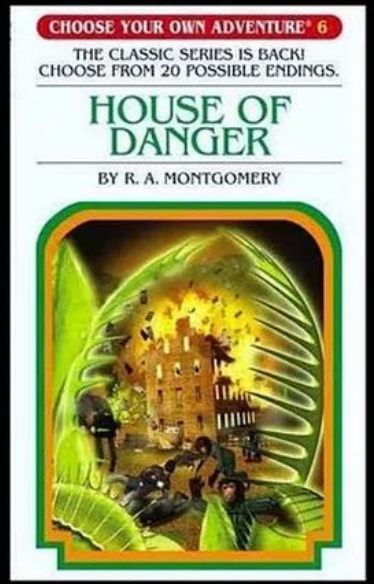
Abstract

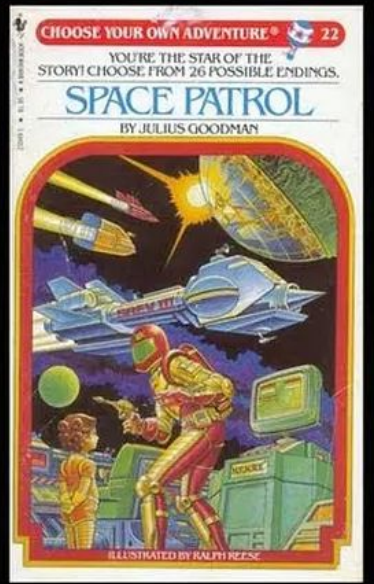
STAC 101 covers the essentials of the SpatioTemporal Asset Catalog, with practical tips for users and data providers to publish, find, and use geospatial data effectively.

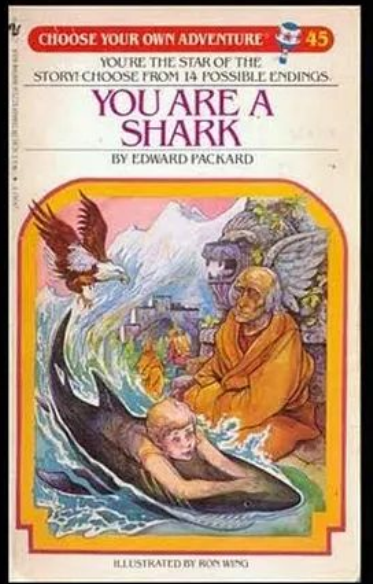
The SpatioTemporal Asset Catalog (STAC) is an open standard that's reshaping how we find, share, and use geospatial and Earth observation data. This session gives new users and data providers the practical essentials for getting started with STAC and using it well.

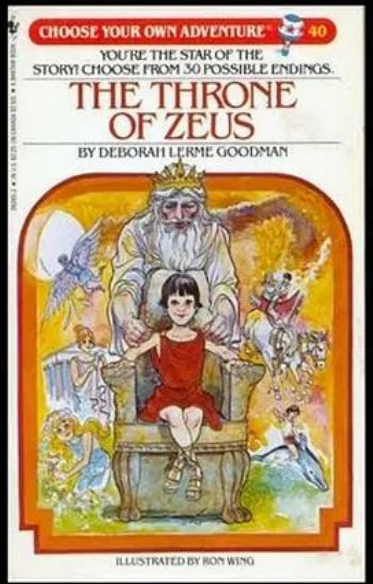
We'll break down the core concepts—Catalogs, Collections, Items, and Extensions—and explain

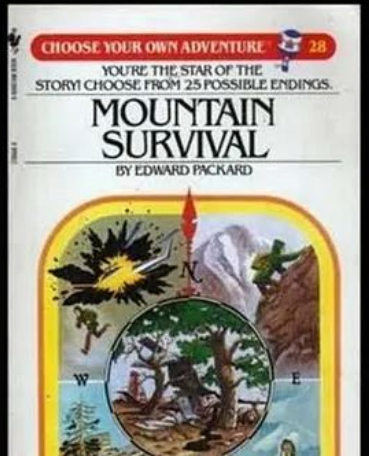

CHOOSE YOUR OWN ADVENTURE • 6
YOU'RE THE HERO OF THE STORY! CHOOSE FROM 40 POSSIBLE ENDINGS.
YOUR CODE NAME IS JONAH
BY EDWARD PACKARD
ILLUSTRATED BY PAUL GRANGER


CHOOSE YOUR OWN ADVENTURE® 6
THE CLASSIC SERIES IS BACK! CHOOSE FROM 20 POSSIBLE ENDINGS.
HOUSE OF DANGER
BY R. A. MONTGOMERY


CHOOSE YOUR OWN ADVENTURE® 22
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 26 POSSIBLE ENDINGS.
SPACE PATROL
BY JULIUS GOODMAN
ILLUSTRATED BY RALPH REESE


CHOOSE YOUR OWN ADVENTURE® 45
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 14 POSSIBLE ENDINGS.
YOU ARE A SHARK
BY EDWARD PACKARD
ILLUSTRATED BY RON WING


CHOOSE YOUR OWN ADVENTURE® 40
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 30 POSSIBLE ENDINGS.
THE THRONE OF ZEUS
BY DEBORAH LERME GOODMAN
ILLUSTRATED BY RON WING


CHOOSE YOUR OWN ADVENTURE® 28
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 25 POSSIBLE ENDINGS.
MOUNTAIN SURVIVAL
BY EDWARD PACKARD
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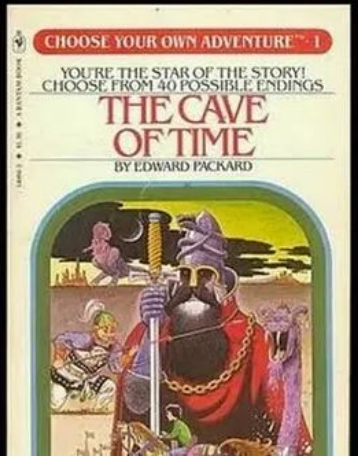
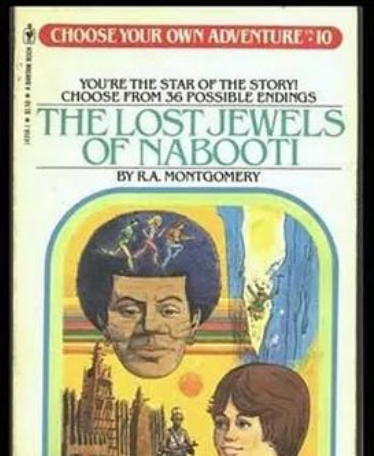
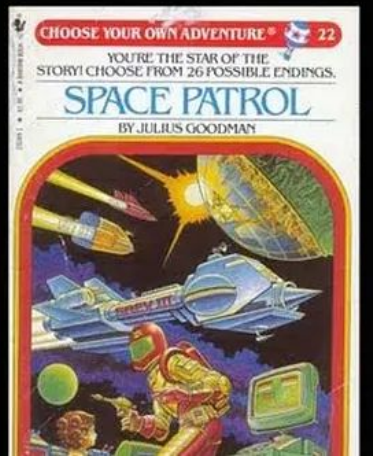

CHOOSE YOUR OWN ADVENTURE™ 1
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 40 POSSIBLE ENDINGS.
THE CAVE OF TIME
BY EDWARD PACKARD


CHOOSE YOUR OWN ADVENTURE® 10
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 36 POSSIBLE ENDINGS
THE LOST JEWELS OF NABOOTI
BY R.A. MONTGOMERY


CHOOSE YOUR OWN ADVENTURE® 22
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 26 POSSIBLE ENDINGS.
SPACE PATROL
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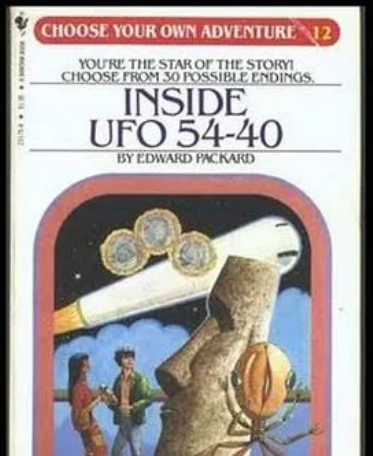
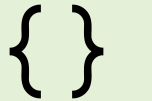

CHOOSE YOUR OWN ADVENTURE® 12
YOU'RE THE STAR OF THE STORY! CHOOSE FROM 30 POSSIBLE ENDINGS.
INSIDE UFO 54-40
BY EDWARD PACKARD

What is STAC

STAC is a suite of specifications and open-source ecosystem for describing and searching geospatial data



stac-spec

Metadata language for describing geospatial data

Extensions

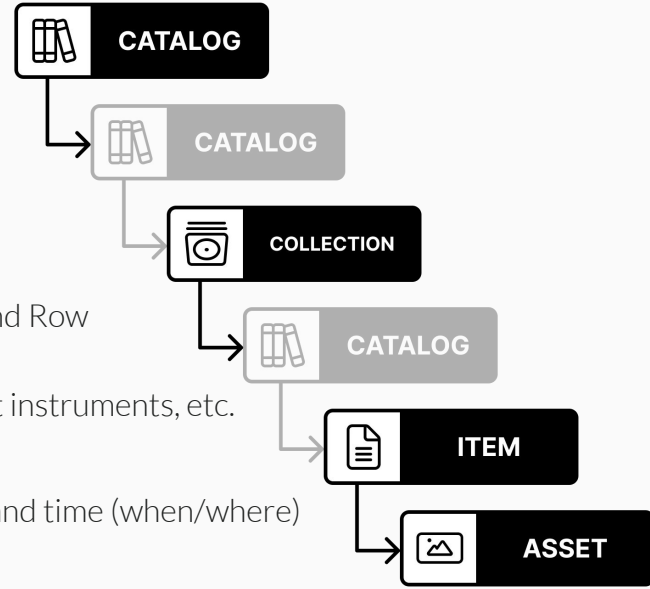
stac-api-spec

Searching and accessing metadata through an API

Extensions

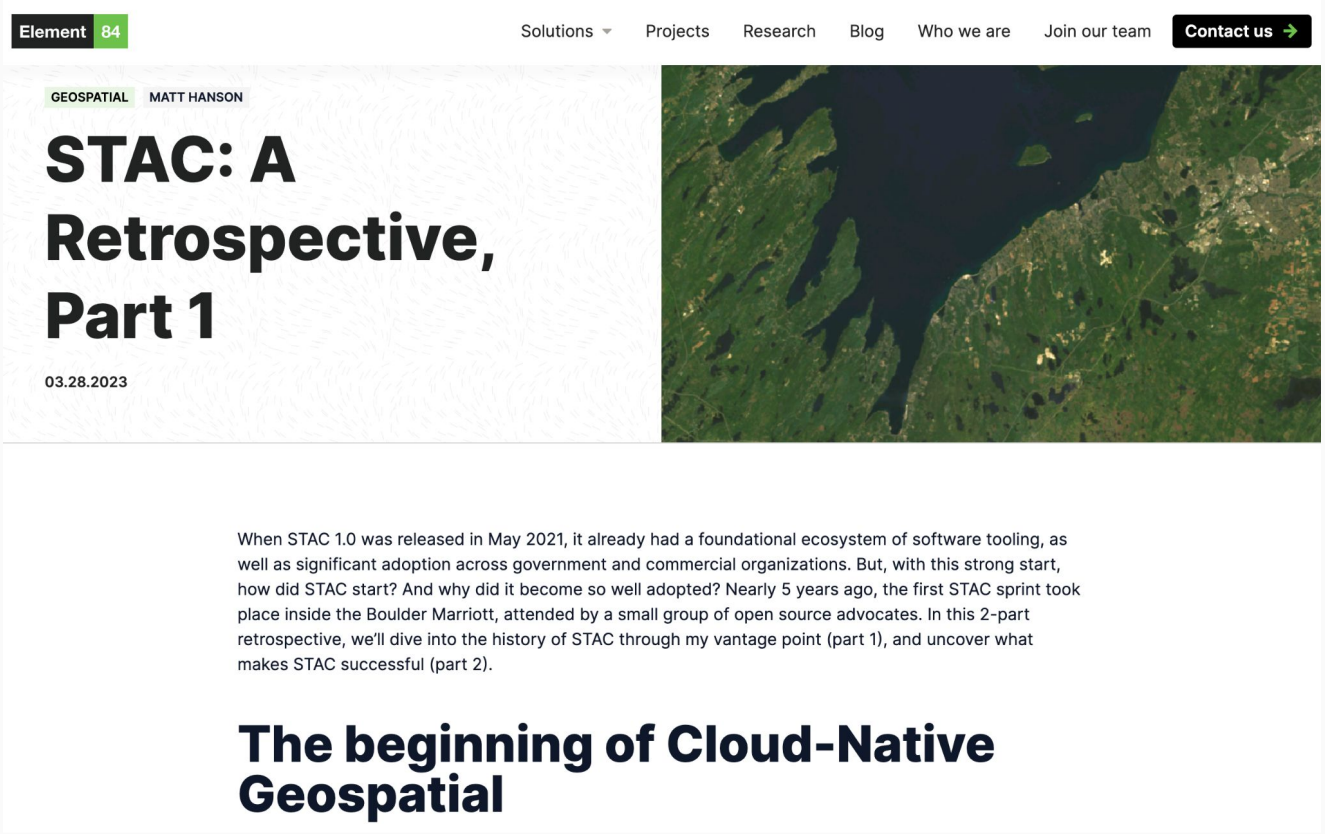
SpatioTemporal Asset Catalogs (STAC)

- Catalog
  - Container that links to other catalogs and collections
  - Top-level catalog is the Root Catalog
- Collection
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Item
  - A single scene or set of datafiles for a specific location at a specific date and time (when/where)
- Asset
  - An individual data file



Key Points

- Built on standards (JSON, GeoJSON, OGC APIs)
- Simple and stable core structure
- Metadata decoupled from data
- Supported by an Open-Source ecosystem



Element 84

https://www.element84.com/geospatial/stac-a-retrospective-part-1/

STAC: A Retrospective, Part 2 (Why STAC was Successful)

10.28.2025

I. The Revolution Wasn't Televised

The way we access and analyze geospatial data has shifted so fundamentally that it's easy to lose perspective on what things were like before. Less than a decade ago, accessing and analyzing geospatial data was a fractured affair; a hard-to-navigate landscape of bespoke APIs and custom tooling. Every new dataset, whether from a government agency or a commercial provider, was its own puzzle. You couldn't just find the data; you had to reverse-engineer the means to use it. Today, nearly eight years after the first STAC sprint, we have an interoperable ecosystem built around a common language for describing geospatial information: the SpatioTemporal Asset Catalog (STAC).

Element 84

https://element84.com/software-engineering/stac-a-retrospective-part-2-why-stac-was-successful/

The Quest for Value

Dive into the chaos and discover why a common standard was essential

Begin

The Library

Choose the form your catalog takes: static files, dynamic services, or parquet-powered

Enter

The Beastiary

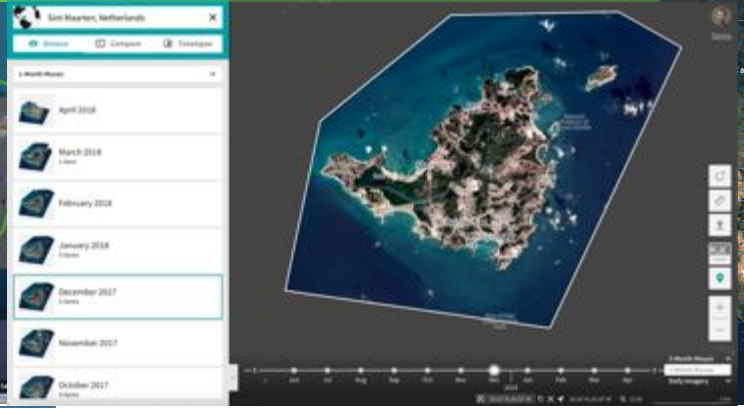
Learn of the different entities within the world of STAC

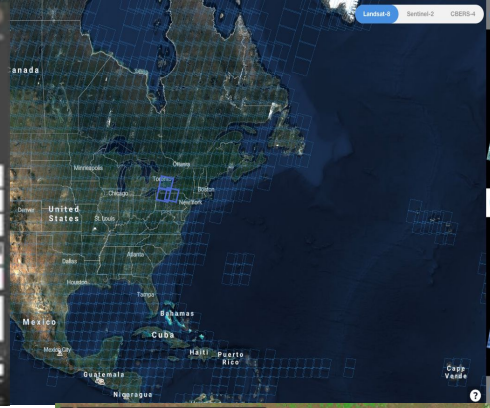
Enter

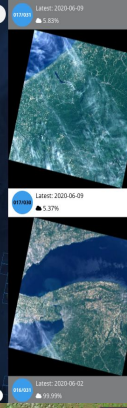
Explore the Cave

Why STAC Matters

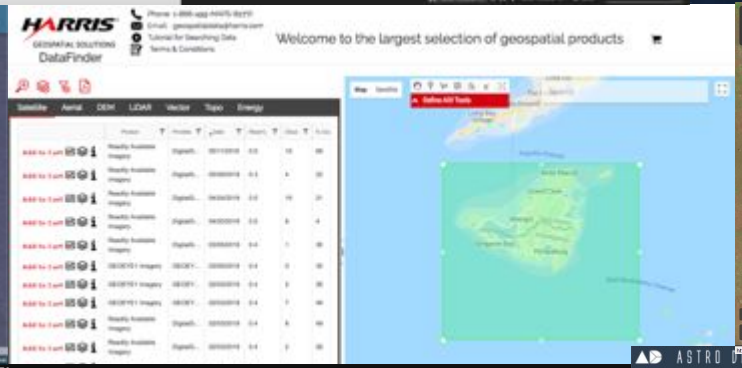
Saint Martin
Area of Interest
Filters

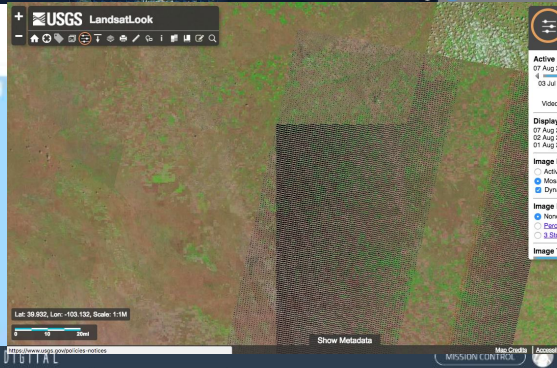

Sint Maarten, Netherlands
Browse
Compare
Timelapse
April 2018
March 2018
February 2018
January 2018
December 2017
November 2017
October 2017

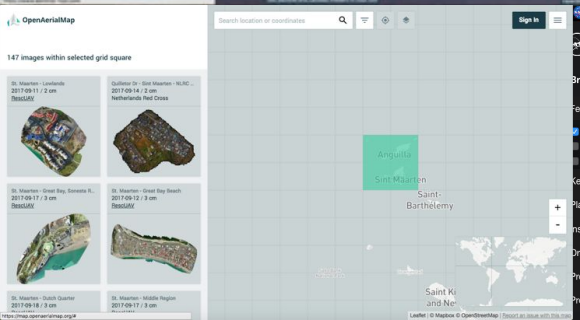

Landsat-8
Sentinel-2
CBERS-4
United States
Mexico
Cuba
Bahamas
Haiti
Puerto Rico
Guatemala
Nicaragua

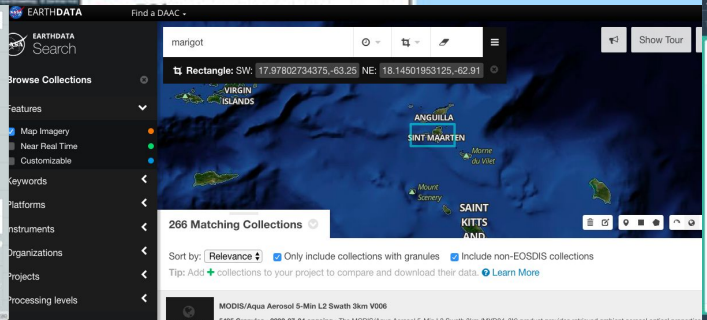


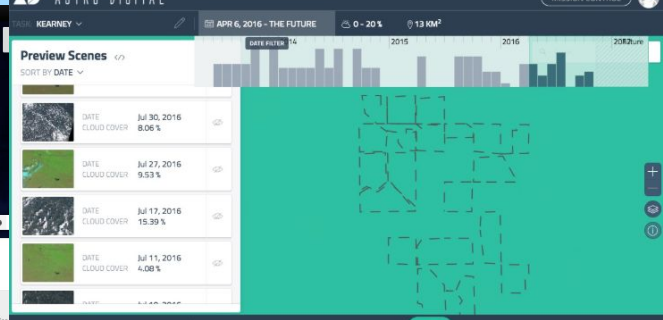
SENTINEL Hub Playground
Rendering
Effects
Natural color

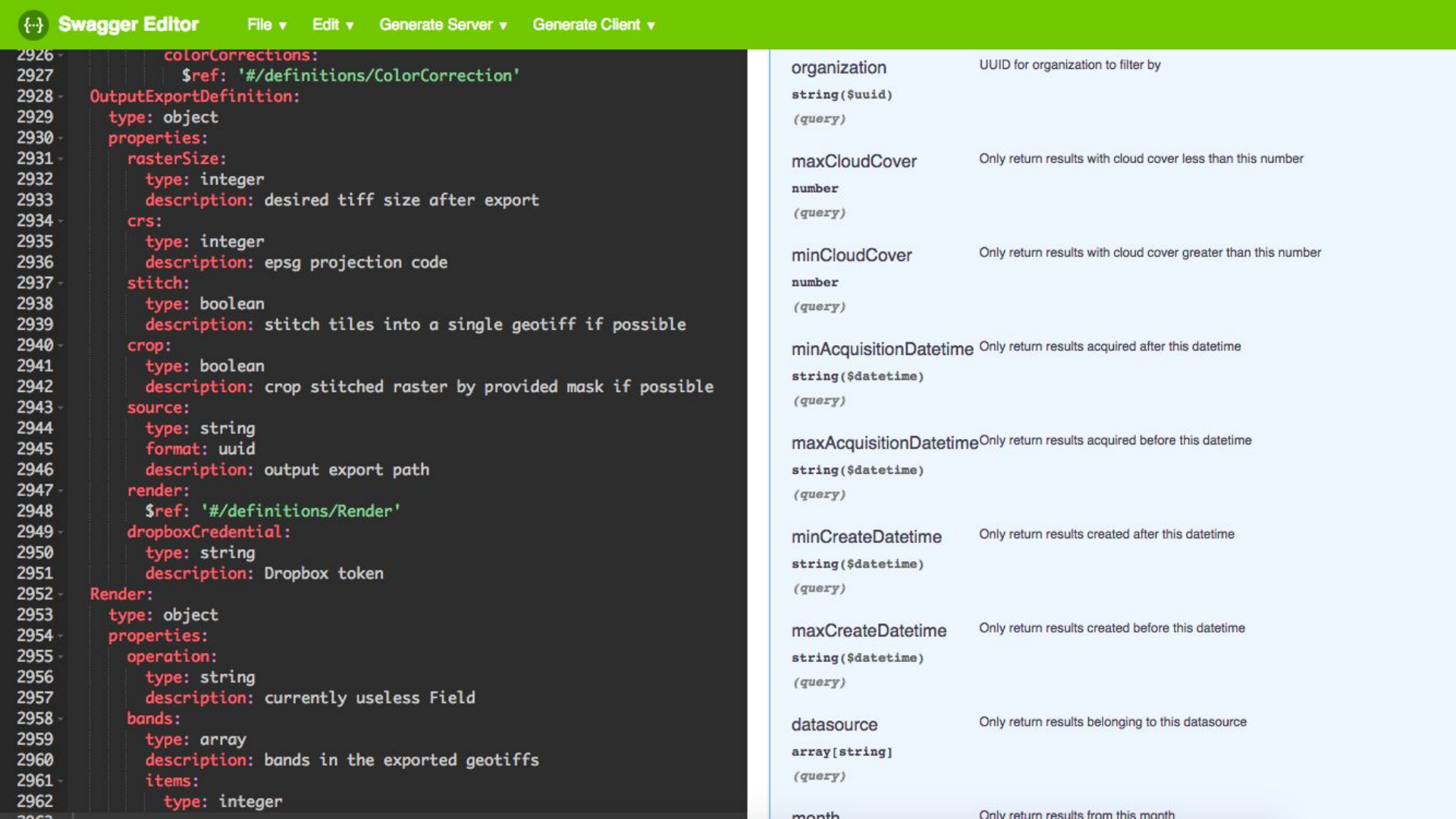

HARRIS GEOSPATIAL SOLUTIONS
DataFinder
Welcome to the largest selection of geospatial products
Satellite
Aerial
DEM
LiDAR
Vector
Topo
Energy


USGS LandsatLook
Lat: 39.932, Lon: -103.132, Scale: 1:1M
Show Metadata

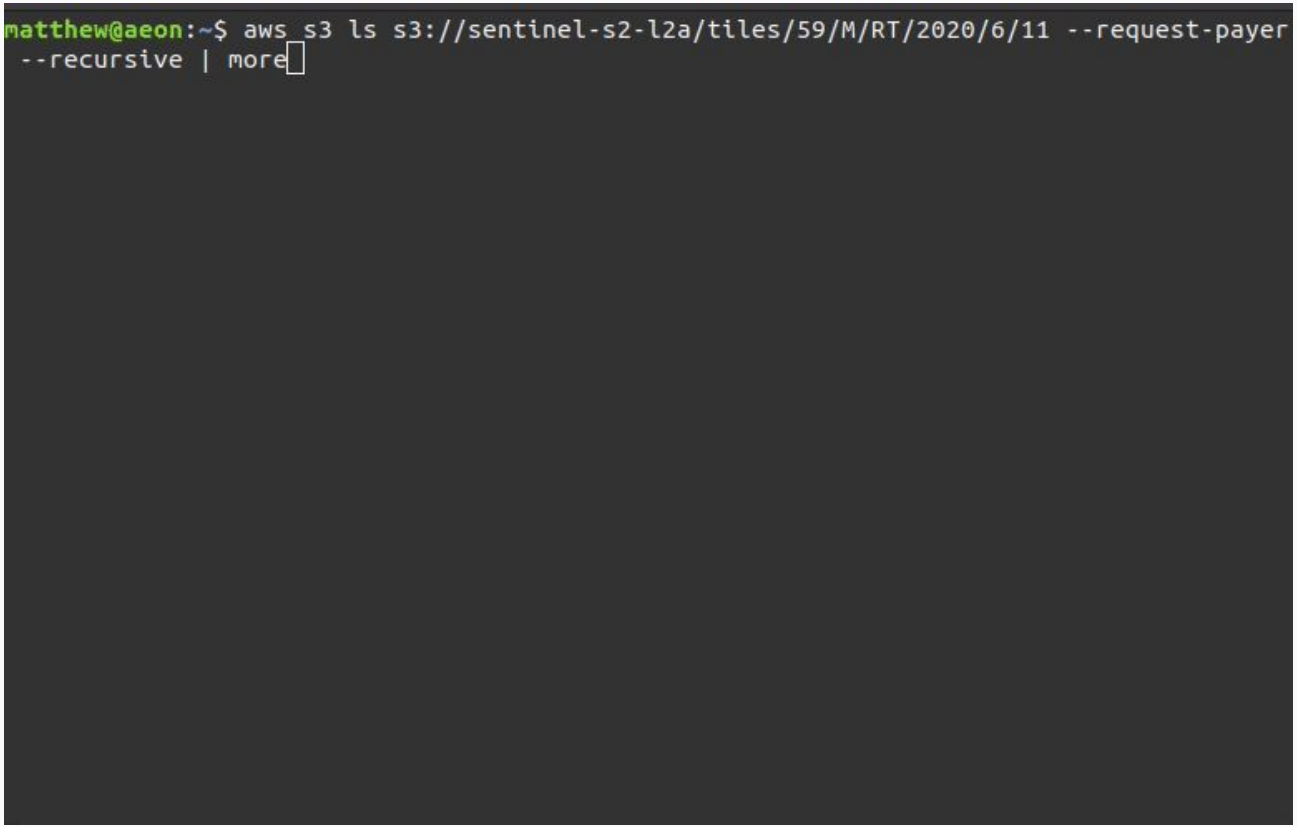

OpenAerialMap
147 images within selected grid square
Anguilla
Sint Maarten
Saint-Barthelemy


EARTHDATA
Search
Find a DAAC
Browse Collections
Features
Map Imagery
Near Real Time
Customizable
Keywords
Platforms
Instruments
Organizations
Projects
Processing levels
marigot
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
Show Tour
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
Tip: Add collections to your project to compare and download their data. Learn More
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006

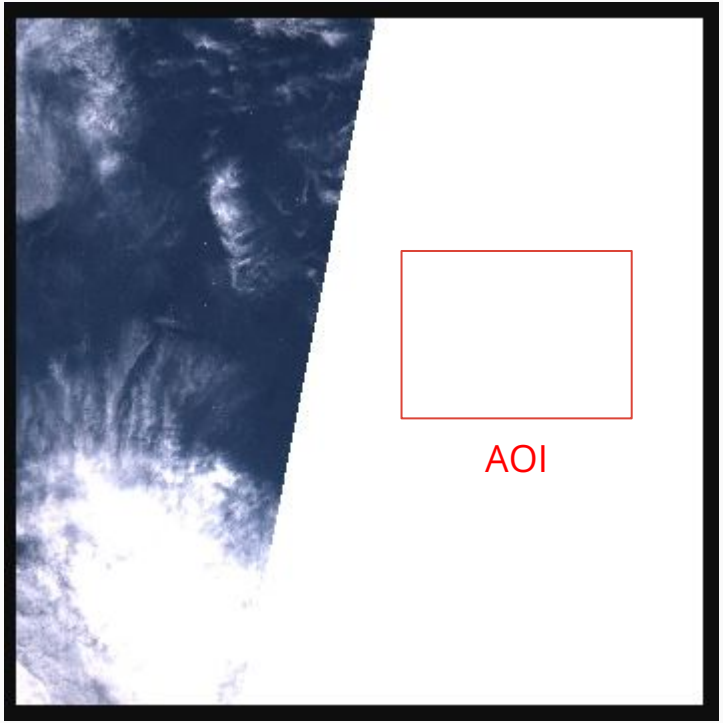

ASTRO DIGITAL
MISSION CONTROL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
Jul 30, 2016
8.06 %
Jul 27, 2016
9.93 %
Jul 17, 2016
15.39 %
Jul 11, 2016
4.08 %


Swagger Editor
File
Edit
Generate Server
Generate Client
colorCorrections:
$ref: '#/definitions/ColorCorrection'
OutputExportDefinition:
type: object
properties:
rasterSize:
type: integer
description: desired tiff size after export
crs:
type: integer
description: epsg projection code
stitch:
type: boolean
description: stitch tiles into a single geotiff if possible
crop:
type: boolean
description: crop stitched raster by provided mask if possible
source:
type: string
format: uuid
description: output export path
render:
$ref: '#/definitions/Render'
dropboxCredential:
type: string
description: Dropbox token
Render:
type: object
properties:
operation:
type: string
description: currently useless Field
bands:
type: array
description: bands in the exported geotiffs
items:
type: integer
organization
UUID for organization to filter by
string($uuid)
(query)
maxCloudCover
Only return results with cloud cover less than this number
number
(query)
minCloudCover
Only return results with cloud cover greater than this number
number
(query)
minAcquisitionDatetime
Only return results acquired after this datetime
string($datetime)
(query)
maxAcquisitionDatetime
Only return results acquired before this datetime
string($datetime)
(query)
minCreateDatetime
Only return results created after this datetime
string($datetime)
(query)
maxCreateDatetime
Only return results created before this datetime
string($datetime)
(query)
datasource
Only return results belonging to this datasource
array[string]
(query)

Discovery via tile



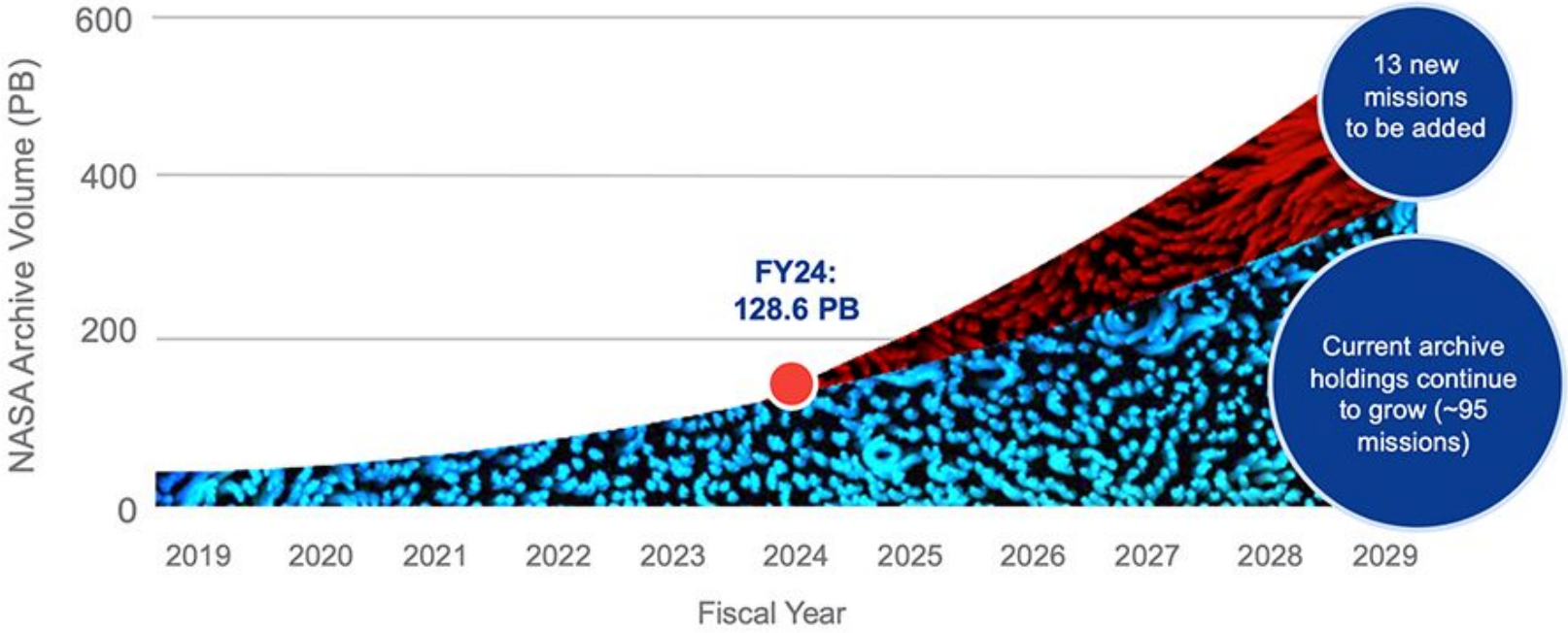
Tile Geometry vs Data Geometry



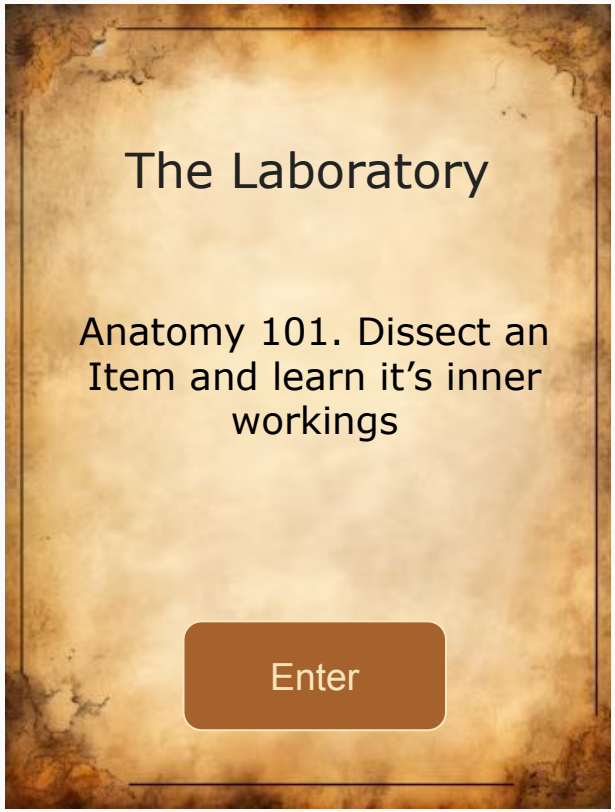
Open Geospatial Consortium (OGC) Standards

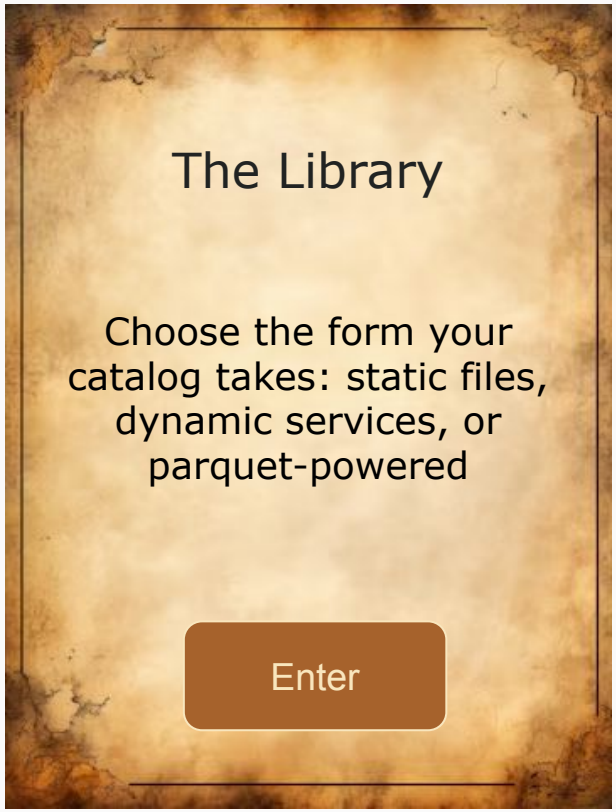
OGC API (New)	OGC Legacy (Old)	What is it for?
Common	OWS Common	Common API elements
Features	Web Feature Service (WFS)	Serving geospatial vector data
Coverages	Web Coverage Services (WCS)	Serving geospatial raster data
Maps	Web Map Service (WMS)	Serving rendered map layers
Tiles	Web Map Tile Service (WMTS)	Serving data as tiles
Processes	Web Processing Service (WPS)	Invoking geospatial processing requests
Records	Catalog Service for the Web (CSW)	Searching metadata

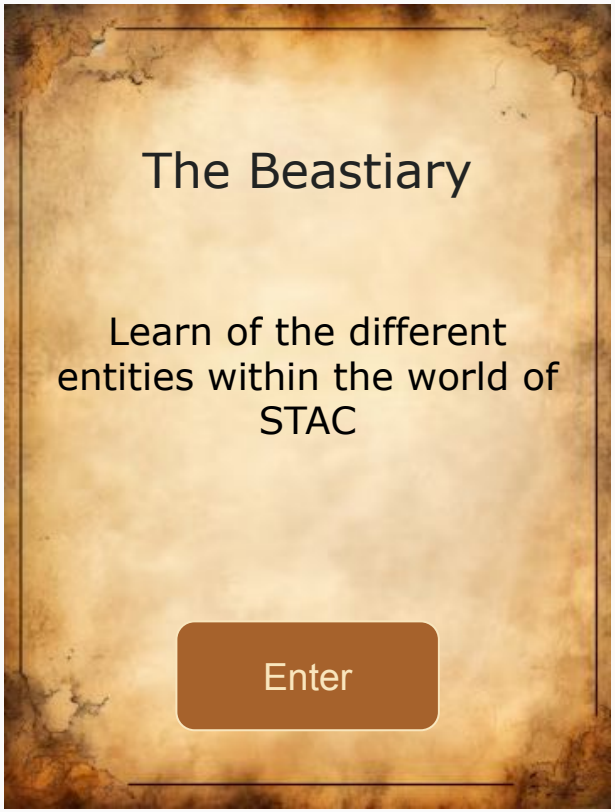
Earth Science Data Archive Growth Projection



If your metadata is not indexed
your data does not exist



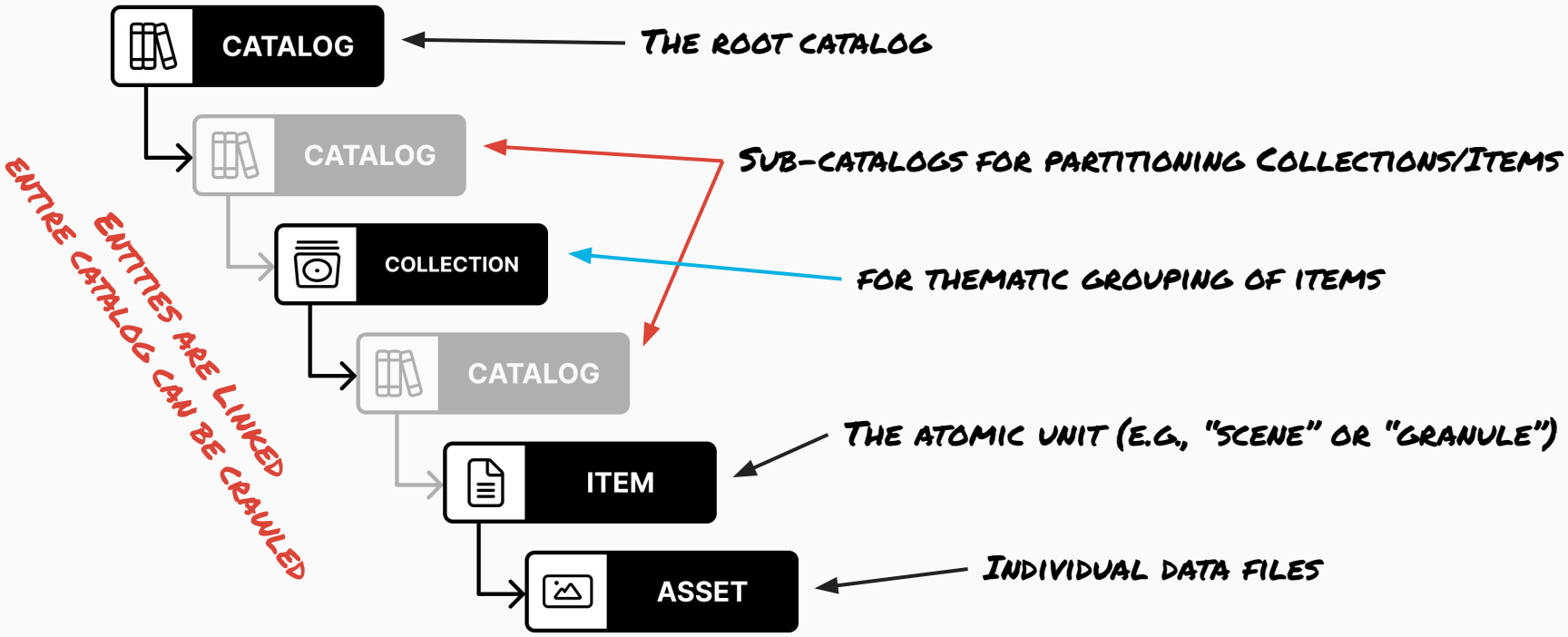




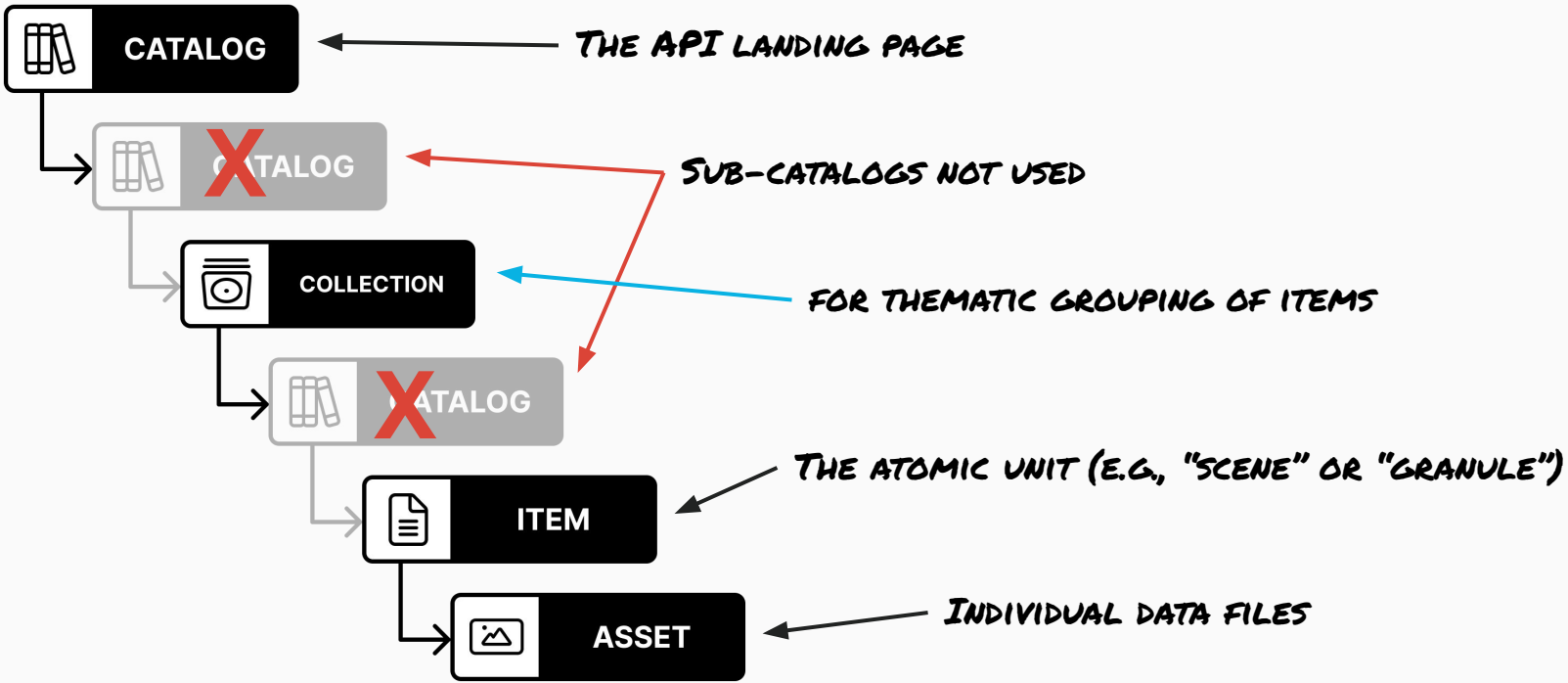
Explore the Cave

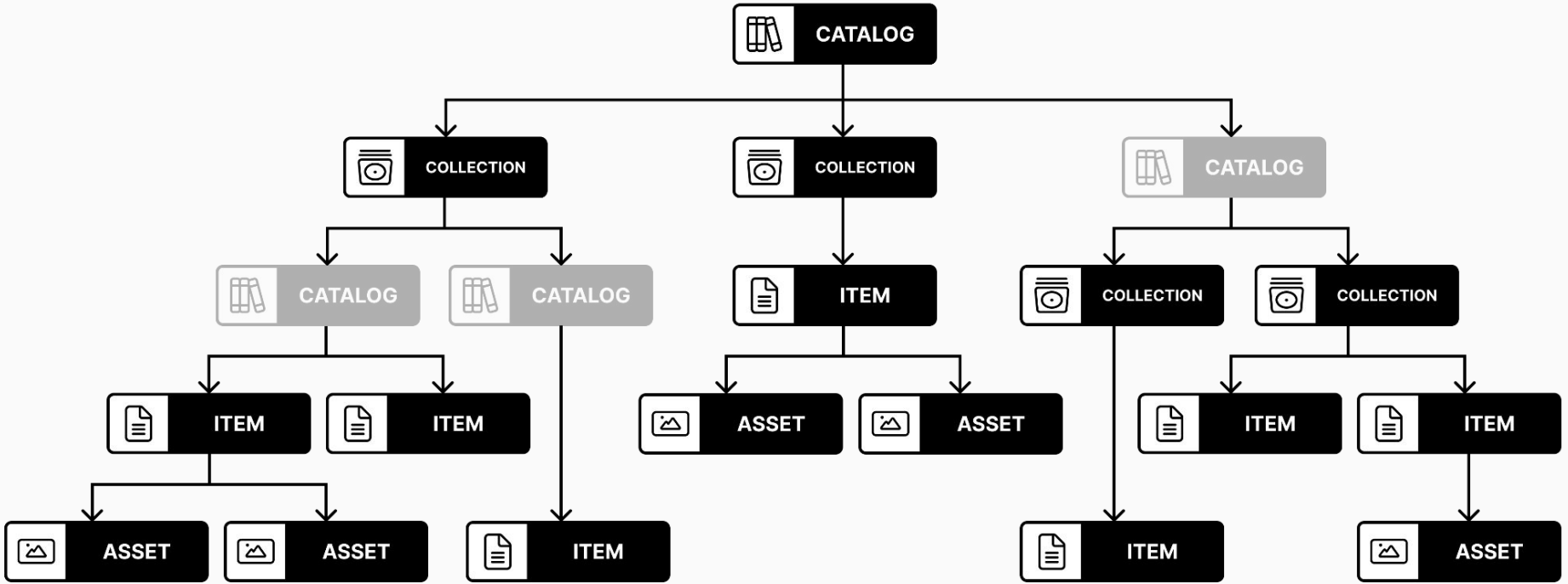
Types of Catalogs

Static Catalogs



“Dynamic” Catalog (an API)




CATALOG
COLLECTION
COLLECTION
CATALOG
CATALOG
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ITEM
COLLECTION
COLLECTION
ITEM
ITEM
ASSET
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ITEM
ITEM
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ASSET

Introduction to STAC GeoParquet

Radiant Earth · Follow

Published in Radiant Earth Insights · 6 min read · Aug 30, 2024

9 1

By Tom Augspurger, Kyle Barron, and Chris Holmes

This post introduces STAC GeoParquet, a specification and library for storing and serving SpatioTemporal Asset Catalogs (STAC) metadata as GeoParquet. By building on GeoParquet, STAC GeoParquet makes it easy to store, transmit, and analyze large collections of STAC items. It makes for a nice complement to a STAC API.

https://medium.com/radiant-earth-insights/introduction-to-stac-geoparquet-031a7902e134

STAC-GeoParquet

Convert STAC items between JSON, GeoParquet, pgstac, and Delta Lake.

Purpose

The STAC spec defines a JSON-based schema. But it can be hard to manage and search through many millions of STAC items in JSON format. For one, JSON is very large on disk. And you need to parse the entire JSON data into memory to extract just a small piece of information, say the `datetime` and one `asset` of an Item.

GeoParquet can be a good complement to JSON for many bulk-access and analytic use cases. While STAC Items are commonly distributed as individual JSON files on object storage or through a STAC API, STAC GeoParquet allows users to access a large number of STAC items in bulk without making repeated HTTP requests.

For analytic questions like "find the items in the Sentinel-2 collection in June 2024 over New York City with cloud cover of less than 20%" it can be much, much faster to find the relevant data from a GeoParquet source than from JSON, because GeoParquet needs to load only the relevant columns for that query, not the full data.

See the STAC-GeoParquet specification for details on the exact schema of the written Parquet files.

https://github.com/stac-utils/stac-geoparquet

The Laboratory

Anatomy 101. Dissect an Item and learn it's inner workings

Enter

The Courtyard

Explore metadata allowed to freely move where it's most useful: Item, Asset, or Band

Enter

The Beastiary

Learn of the different entities within the world of STAC

Enter

Explore the Cave

STAC Entities

Catalog

Element	Type	Description
type	string	**REQUIRED.** Set to `Catalog` if this Catalog only implements the Catalog spec.
stac_version	string	**REQUIRED.** The STAC version the Catalog implements.
stac_extensions	[string]	A list of extension identifiers the Catalog implements.
id	string	**REQUIRED.** Identifier for the Catalog.
title	string	A short descriptive one-line title for the Catalog.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the Catalog. CommonMark 0.29 syntax MAY be used for rich text representation.
links	[Link Object]	**REQUIRED.** A list of references to other documents.

Link Object

This object describes a relationship with another entity. Data providers are advised to be liberal with links.

Field Name	Type	Description
href	string	**REQUIRED.** The actual link in the format of an URL. Relative and absolute links are both allowed.
rel	string	**REQUIRED.** Relationship between the current document and the linked document. See chapter "Relation types" for more information.
type	string	Media type of the referenced entity.
title	string	A human readable title to be used in rendered displays of the link.

Element 84

Collection

Element	Type	Description
type	string	**REQUIRED.** Must be set to `Collection` to be a valid Collection.
stac_version	string	**REQUIRED.** The STAC version the Collection implements.
stac_extensions	[string]	A list of extension identifiers the Collection implements.
id	string	**REQUIRED.** Identifier for the Collection that is unique across the provider.
title	string	A short descriptive one-line title for the Collection.
description	string	**REQUIRED.** Detailed multi-line description to fully explain the Collection. CommonMark 0.29 syntax MAY be used for rich text representation.
keywords	[string]	List of keywords describing the Collection.
license	string	**REQUIRED.** Collection's license(s), either a SPDX License identifier, `various` if multiple licenses apply or `proprietary` for all other cases.
providers	[Provider Object]	A list of providers, which may include all organizations capturing or processing the data or the hosting provider. Providers should be listed in chronological order with the most recent provider being the last element of the list.
extent	Extent Object	**REQUIRED.** Spatial and temporal extents.
summaries	Map<string, [*]|Range Object|JSON Schema Object>	STRONGLY RECOMMENDED. A map of property summaries, either a set of values, a range of values or a JSON Schema.
links	[Link Object]	**REQUIRED.** A list of references to other documents.
assets	Map<string, Asset Object>	Dictionary of asset objects that can be downloaded, each with a unique key.

Element 84

How should I organize my Collections?

- Consider collections from the user
- Uniform assets

Core STAC Item

Field Name	Type	Description
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. The ID should be unique within the Collection that contains the Item.
geometry	GeoJSON Geometry Object | null	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED if `geometry` is not `null`.** Bounding Box of the asset represented by this Item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the Item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present and is *not allowed* otherwise. This field provides an easy way for a user to search for any Items that belong in a specified Collection. Must be a non-empty string.

The Laboratory

Anatomy 101. Dissect an Item and learn it's inner workings

Enter

The Courtyard

Explore metadata allowed to freely move where it's most useful: Item, Asset, or Band

Enter

Discovery Quest

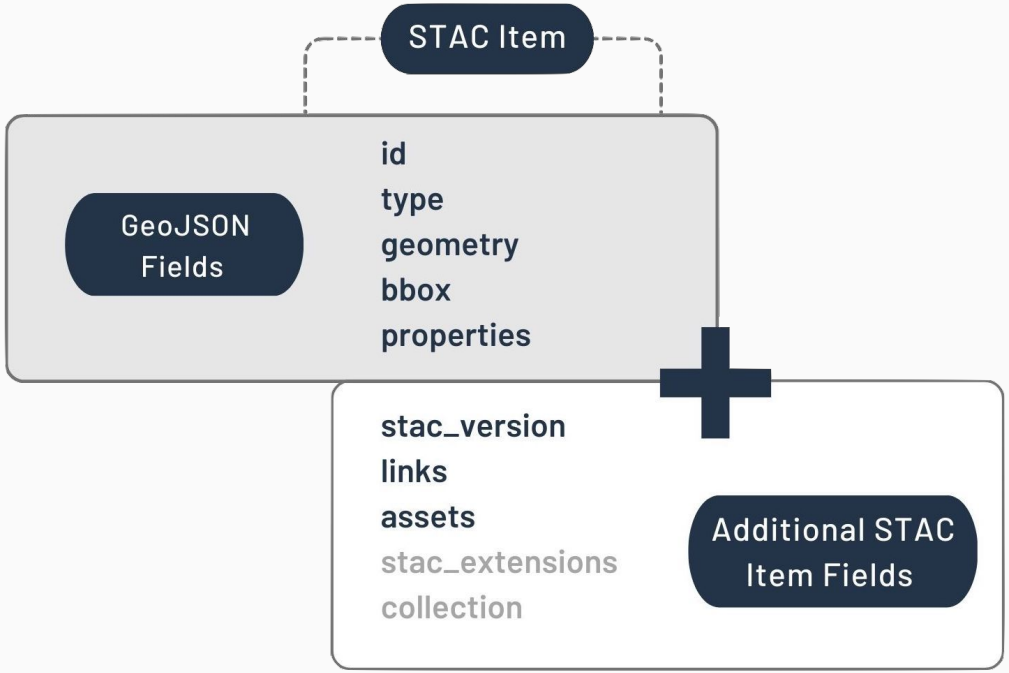
Open the door to a unified landscape - searchable, filterable, and alive.

Enter

Explore the Cave

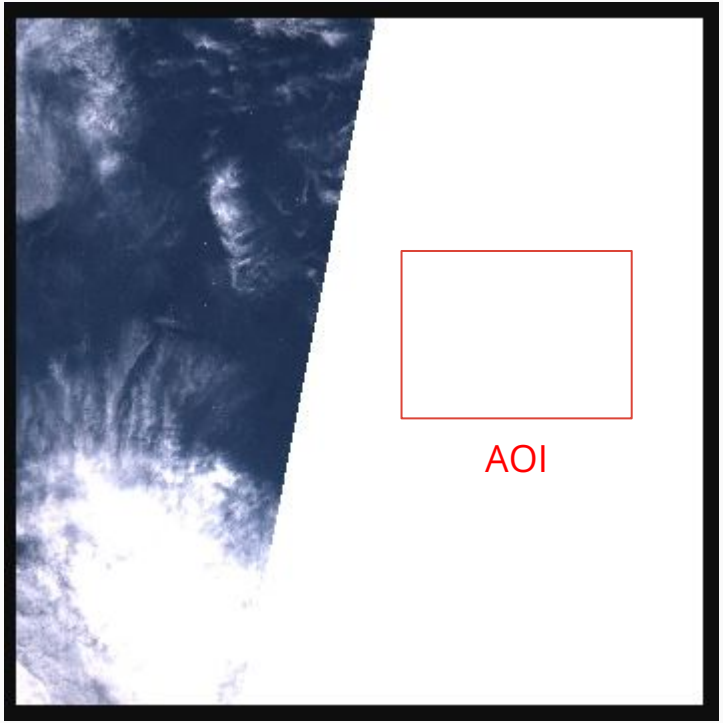
Anatomy of an Item

The Atomic Unit - an Item

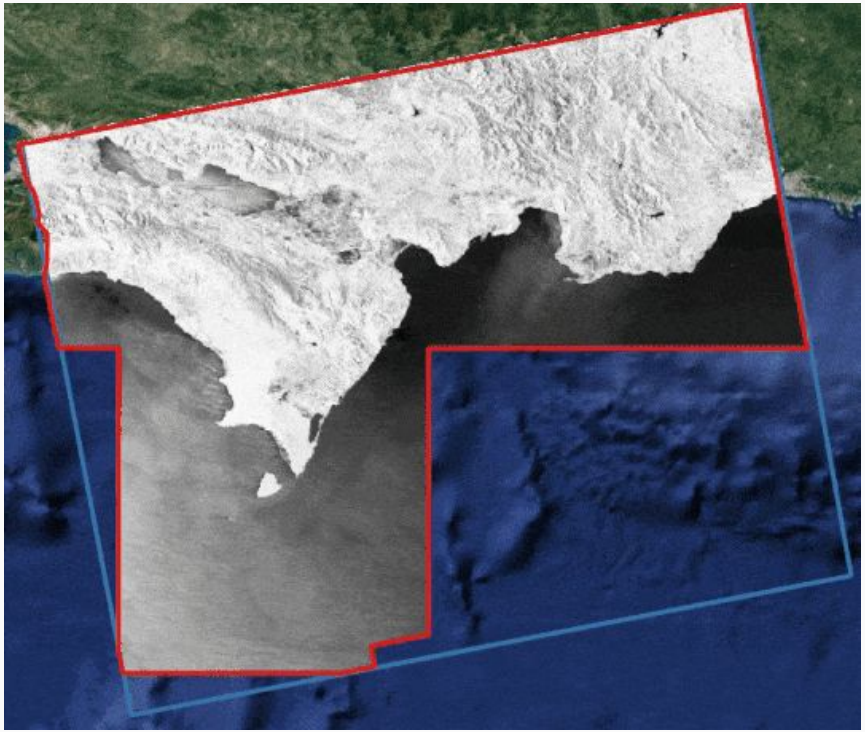



```
{
  "type": "Feature",
  "stac_version": "1.0.0",
  "id": "S2A_T22JDM_20240508T132235_L2A",
  "properties": {…
  },
  "geometry": {
    "type": "Polygon",
    "coordinates": [
      [
        [-51.64997287741106, -29.82887736130735],
        [-51.93692244077125, -30.817923786828445],
        [-50.897965022919415, -30.821271019233876],
        [-50.89898692183221, -29.83043772553916],
        [-51.64997287741106, -29.82887736130735]
      ]
    ]
  },
  "links": […
  ],
  "assets": {…
  },
  "bbox": [-51.936922, -30.821271, -50.897965, -29.828877],
  "stac_extensions": […
  ],
  "collection": "sentinel-2-c1-l2a"
}
```

Tile Geometry vs Data Geometry



Footprint Example - Sentinel-1



Element 84

See https://element84.com/blog/the-stactools-raster-footprint-utility

Links


```
"links": [
  {
    "rel": "self",
    "type": "application/geo+json",
    "href": "https://earth-search.aws.element84.com/v1/collections/sentinel-2-c1-l2a/items/S2A_T22JDM_20240508T132235_L2A"
  },
  {
    "rel": "canonical",
    "href": "s3://e84-earth-search-sentinel-data/sentinel-2-c1-l2a/22/J/DM/2024/5/S2A_T22JDM_20240508T132235_L2A/S2A_T22JDM_20240508T132235_L2A.json",
    "type": "application/json"
  },
  {
    "rel": "via",
    "href": "s3://sentinel-s2-l2a/tiles/22/J/DM/2024/5/8/0/metadata.xml",
    "type": "application/xml",
    "title": "Granule Metadata in Sinergize RODA Archive"
  },
  {
    "rel": "parent",
    "type": "application/json",
    "href": "https://earth-search.aws.element84.com/v1/collections/sentinel-2-c1-l2a"
  },
  {
    "rel": "collection",
    "type": "application/json",
    "href": "https://earth-search.aws.element84.com/v1/collections/sentinel-2-c1-l2a"
  },
  {
    "rel": "root",
    "type": "application/json",
    "href": "https://earth-search.aws.element84.com/v1"
  },
```

Item Properties


```
"id": "S2A_T22JDM_20240508T132235_L2A",
"properties": {
  "title": "Sentinel-2 Scene",
  "datetime": "2024-05-08T13:30:53.463000Z",
  "created": "2024-05-08T22:20:35.704Z",
  "updated": "2024-05-08T22:20:35.704Z",
  "platform": "sentinel-2a",
  "constellation": "sentinel-2",
  "instruments": ["msi"]
},
"geometry": {
  "type": "Polygon",
```

Assets

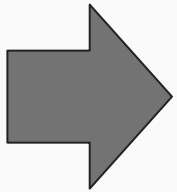

```
"assets": {
  "red": {
    "href": "https://e84-earth-search-sentinel-data.s3.us-west-2.amazonaws.com/sentinel-2-c1-l2a/22/J/DM/2024/5/S2A_T22JDM_20240508T132235_L2A/B04.tif",
    "type": "image/tiff; application=geotiff; profile=cloud-optimized",
    "title": "Red - 10m",
    "name": "B04",
    "nodata": 0,
    "data_type": "uint16",
    "gsd": 10,
    "roles": ["data", "reflectance"]
  },
  "green": {
    "href": "https://e84-earth-search-sentinel-data.s3.us-west-2.amazonaws.com/sentinel-2-c1-l2a/22/J/DM/2024/5/S2A_T22JDM_20240508T132235_L2A/B03.tif",
    "type": "image/tiff; application=geotiff; profile=cloud-optimized",
    "title": "Green - 10m",
    "name": "B03",
    "nodata": 0,
    "data_type": "uint16",
    "gsd": 10,
    "roles": ["data", "reflectance"]
  },
  "blue": {
```

Bands


```
"visual": {
  "href": "https://e84-earth-search-sentinel-data.s3.us-west-2.amazonaws.com/sentinel-2-c1-l2a/2
  "type": "image/tiff; application=geotiff; profile=cloud-optimized",
  "title": "True color image",
  "bands": [
    {
      "name": "B04",
      "eo:common_name": "red",
      "eo:center_wavelength": 0.665,
      "eo:full_width_half_max": 0.038,
      "nodata": 0,
      "data_type": "uint8",
      "raster:spatial_resolution": 10
    },
    {
      "name": "B03",
      "eo:common_name": "green",
      "eo:center_wavelength": 0.56,
      "eo:full_width_half_max": 0.045,
      "nodata": 0,
      "data_type": "uint8",
      "raster:spatial_resolution": 10
    },
    {
      "name": "B02",
      "eo:common_name": "blue",
      "eo:center_wavelength": 0.49,
      "eo:full_width_half_max": 0.098,
      "nodata": 0,
      "data_type": "uint8",
      "raster:spatial_resolution": 10
    }
  ],
  "gsd": 10,
  "proj:shape": [10980, 10980],
```

STAC 1.1 Bands

```
{
    "eo:bands": [{
     "name": "B4",
     "description": "Red band",
     "common_name": "red",
     "center_wavelength": 0.665,
     "full_width_half_max": 0.038
    }],
    "raster:bands": [{
     "data_type": "float32",
     "nodata": "nan",
     "statistics":
        {"minimum": 0, "maximum": 1},
     "spatial_resolution": 10
    }]
}
```



```
{
    "bands": [{
        "name": "B4",
        "description": "Red band",
        "data_type": "float32",
        "nodata": "nan",
        "statistics":
            {"minimum": 0, "maximum": 1},
        "eo:common_name": "red",
        "eo:center_wavelength": 0.665,
        "eo:full_width_half_max": 0.038,
        "raster:spatial_resolution": 10
    }]
}
```

The Arboretum

Step into a garden tended by contributors across the world.

Enter

The Courtyard

Explore metadata allowed to freely move where it's most useful: Item, Asset, or Band

Enter

Survival Guide

Use the field rules that prevent hidden traps and dead ends.

Enter

Explore the Cave

The Common Metadata Model

Common Metadata

https://github.com/radiantearth/stac-spec/blob/master/commons/common-metadata.md

Common Metadata

Field Name	Type	Description
title	string	A human readable title describing the STAC entity.
description	string	Detailed multi-line description to fully explain the STAC entity. CommonMark 0.29 syntax MAY be used for rich text representation.
keywords	[string]	List of keywords describing the STAC entity.
roles	[string]	The semantic roles of the entity, e.g. for assets, links, providers, bands, etc.

Field Name	Type	Description
platform	string	Unique name of the specific platform to which the instrument is attached.
instruments	[string]	Name of instrument or sensor used (e.g., MODIS, ASTER, OLI, Canon F-1).
constellation	string	Name of the constellation to which the platform belongs.
mission	string	Name of the mission for which data is collected.
gsd	number	Ground Sample Distance at the sensor, in meters (m), must be greater than 0.

Field Name	Type	Description
nodata	number|string	Value used to identify no-data, see below.
data_type	string	The data type of the values, see below.
statistics	Statistics Object	Statistics of all the values.
unit	string	Unit of measurement of the value, see below.

Element 84

Datetimes

datetime	string|null	See the Item Specification Fields for more information.
created	string	Creation date and time of the corresponding STAC entity or Asset (see below), in UTC.
updated	string	Date and time the corresponding STAC entity or Asset (see below) was updated last, in UTC.

start_datetime	string	The first or start date and time for the resource, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.
end_datetime	string	The last or end date and time for the resource, in UTC. It is formatted as `date-time` according to RFC 3339, section 5.6.

Licensing

Information about the license(s) of the data, which is not necessarily the same license that applies to the metadata. **Licensing information should be defined at the Collection level if possible.**

- JSON Schema

Field Name	Type	Description
license	string	License(s) of the data as SPDX License identifier, SPDX License expression, or `other` (see below).

license: License(s) of the data that the STAC entity provides.

The license(s) can be provided as:

1. SPDX License identifier
2. SPDX License expression
3. String with the value `other` if the license is not on the SPDX license list. The strings `various` and `proprietary` are **deprecated**.

If the license is **not** an SPDX license identifier, links to the license texts SHOULD be added. The links MUST use the `license` link relation type. If there is no public license URL available, it is RECOMMENDED to supplement the STAC Item with the license text in a separate file and link to this file. If no link to a license is included and the `license` field is set to `other` (or one of the deprecated values), the data is private, and consumers have not been granted any explicit right to use it.

License relation

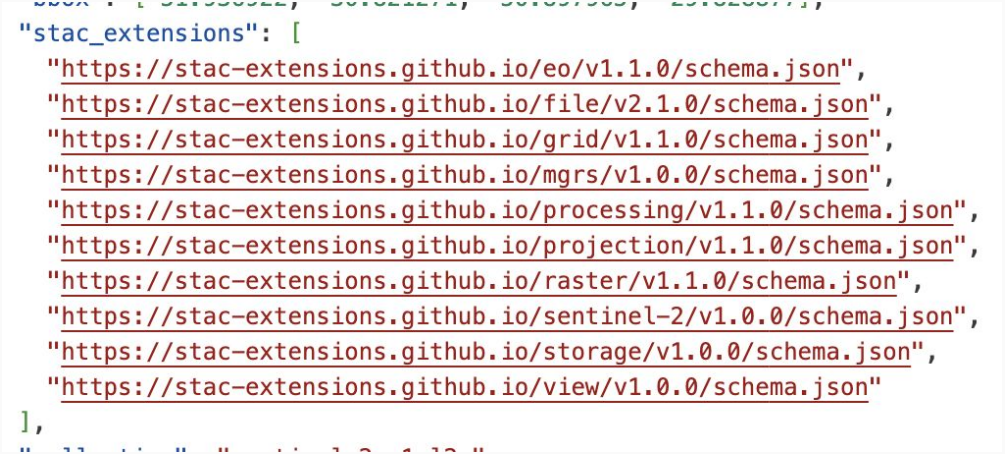
Type	Description
license	The license URL(s) for the resource SHOULD be specified if the `license` field is **not** a SPDX license identifier.

Provider Object

The object provides information about a provider. A provider is any of the organizations that captures or processes the content of the assets and therefore influences the data offered by the STAC implementation. May also include information about the final storage provider hosting the data.

Field Name	Type	Description
name	string	**REQUIRED.** The name of the organization or the individual.
description	string	Multi-line description to add further provider information such as processing details for processors and producers, hosting details for hosts or basic contact information. CommonMark 0.29 syntax MAY be used for rich text representation.
roles	[string]	Roles of the provider. Any of `licensor`, `producer`, `processor` or `host`.
url	string	Homepage on which the provider describes the dataset and publishes contact information.

What about Extensions?



Extensions?


```
"id": "S2A_T22JDM_20240508T132235_L2A",
"properties": {
  "title": "Sentinel-2 Scene",
  "datetime": "2024-05-08T13:30:53.463000Z",
  "created": "2024-05-08T22:20:35.704Z",
  "updated": "2024-05-08T22:20:35.704Z",
  "platform": "sentinel-2a",
  "constellation": "sentinel-2",
  "instruments": ["msi"],
  "eo:cloud_cover": 72.339511,
  "proj:epsg": 32722,
  "proj:centroid": { "lat": -30.35119, "lon": -51.34972 },
  "mgrs:utm_zone": 22,
  "mgrs:latitude_band": "J",
  "mgrs:grid_square": "DM",
  "grid:code": "MGRS-22JDM",
  "view:azimuth": 102.132995978735,
  "view:incidence_angle": 8.43961873489662,
  "view:sun_azimuth": 33.1498997205534,
  "view:sun_elevation": 35.3222396194671,
  "storage:platform": "AWS",
  "storage:region": "us-west-2",
  "storage:requester_pays": false,
  "processing:software": { "sentinel-2-c1-l2a-to-stac": "v2024.02.01" }
},
"geometry": {
```


EO extension

Projection extension

Mgrs extension

Grid extension

View extension

Storage extension

Processing extension

Metadata Placement

```
{
  "type": "Feature",
  "stac_version": "1.1.0",
  "id": "example-item-001",
  "properties": {
    "datetime": "2025-01-15T10:22:00Z",
    "created": "2025-01-16T08:30:00Z",
    "title": "Example Scene 001",
    "description": "A sample satellite scene covering the area of interest.'
    "gsd": 10
  },
  "assets": {
```

```
"assets": {
  "data": {
    "href": "https://example.com/data-file.tif",
    "type": "image/tiff; application=geotiff",
    "title": "Main Data File",
    "description": "Cloud-free geotiff at 10 m GSD.",
    "created": "2025-01-16T08:30:00Z",
    "bands": [
      { "name": "red",   "description": "Red band." },
      { "name": "green", "description": "Green band." },
      { "name": "blue",  "description": "Blue band." }
    ]
  },
  "overview": {
    "href": "https://example.com/overview.tif",
    "type": "image/tiff; application=geotiff; profile=cloud-optimized",
    "title": "Overview Image",
    "description": "Lower-resolution quicklook image.",
    "gsd": 30
  }
}
```

The Arboretum

Step into a garden tended by contributors across the world.

Enter

Discovery Quest

Open the door to a unified landscape - searchable, filterable, and alive.

Begin

Survival Guide

Use the field rules that prevent hidden traps and dead ends.

Enter

Explore the Cave

STAC API

STAC API

STAC API spec
https://github.com/radiantearth/stac-api-spec

OGC API: Features
+
STAC Extensions

OpenAPI Doc: https://stacspec.org/STAC-ext-api.html

OGC API - Features Endpoints

The core OGC API - Features endpoints are shown below, with details provided in an OpenAPI specification document.

Endpoint	Returns	Description
`/`	JSON	Landing page, links to API capabilities
`/conformance`	JSON	Info about standards to which the API conforms
`/collections`	JSON	Object with a list of Collections contained in the catalog and links
`/collections/{collectionId}`	Collection	Returns single Collection JSON
`/collections/{collectionId}/items`	ItemCollection	GeoJSON FeatureCollection-conformant entity of Items in collection
`/collections/{collectionId}/items/{featureId}`	Item	Returns single Item (GeoJSON Feature)

STAC Endpoints

STAC provides some additional endpoints for the root Catalog itself, as well as the capability to search the Catalog. Note that a STAC API does not need to implement OAFeat, in which case it would only support the endpoints given below. See the OpenAPI specification document.

Endpoint	Returns	Description
`/`	Catalog	Extends `/` from OAFeat to return a full STAC catalog.
`/search`	ItemCollection	Retrieves a group of Items matching the provided search predicates, probably containing search metadata from the `search` extension

Search endpoints

Conformance Classes

Conformance Class Table

Name	Specified in	Conformance URI	Description
STAC API - Core	Core	https://api.stacspec.org/v1.0.0/core	Specifies the STAC Landing page `/`, communicating conformance and available endpoints.
STAC API - Item Search	Item Search	https://api.stacspec.org/v1.0.0/item-search	Enables search of all STAC Item objects on the server, with the STAC `[/search](#stac-api-endpoints)` endpoint.
STAC API - Features	Collections and Features	https://api.stacspec.org/v1.0.0/ogcapi-features	Specifies the use of OGC API - Features to serve STAC Item and Collection objects
STAC API - Collections	Collections and Features	https://api.stacspec.org/v1.0.0/collections	Specifies the use of a subset of STAC API - Features to serve Collection objects

```
{
    "stac_version": "1.0.0",
    "id": "example-stac",
    "title": "A simple STAC API Example",
    "description": "This Catalog aims to demonstrate the a simple landing page",
    "type": "Catalog",
    "conformsTo" : [
        "https://api.stacspec.org/v1.0.0/core",
        "https://api.stacspec.org/v1.0.0/collections",
        "https://api.stacspec.org/v1.0.0/ogcapi-features",
        "https://api.stacspec.org/v1.0.0/item-search",
        "http://www.opengis.net/spec/ogcapi-features-1/1.0/conf/core",
        "http://www.opengis.net/spec/ogcapi-features-1/1.0/conf/oas30",
        "http://www.opengis.net/spec/ogcapi-features-1/1.0/conf/geojson"
    ],
    "links": [
        {
            "rel": "self",
            "type": "application/json",
            "href": "https://stac-api.example.com"
        },
        {
            "rel": "root",
            "type": "application/json",
            "href": "https://stac-api.example.com"
        },
        {
            "rel": "conformance",
            "type": "application/json",
            "href": "https://stac-api.example.com/conformance"
        },
```

STAC API Extensions

Maturity Classification	Min Impl #	Description	Stability
Proposal	0	An idea put forward by a community member to gather feedback	Not stable - breaking changes almost guaranteed as implementers try out the idea.
Pilot	1	Idea is fleshed out, with examples and a JSON schema, and implemented in one or more catalogs. Additional implementations encouraged to help give feedback	Approaching stability - breaking changes are not anticipated but can easily come from additional feedback.
Candidate	3	A number of implementers are using it and are standing behind it as a solid extension. User can generally count on an extension at this maturity level.	Mostly stable, breaking changes require a new version and minor changes are unlikely. The extension has a code owner, designated in its README.
Stable	6	Highest current level of maturity. The community of extension maintainers commits to a STAC review process for any changes, which are not made lightly.	Completely stable, all changes require a new version number and review process.
Deprecated	N/A	A previous extension that has likely been superseded by a newer one or did not work out for some reason.	DO NOT USE, is not supported

Stable Extensions

Query	STAC API - Features, STAC API - Item Search	Stable	1.0.0	The Query Extension adds a query parameter that allows additional filtering based on the properties of Item objects.
Sort	STAC API - Collection Search, STAC API - Features, STAC API - Item Search	Stable	1.1.0	The Sort Extension that allows the user to define the fields by which to sort results.

Candidate Extensions

Collection Transaction	STAC API - Collections	Candidate	1.0.0-beta.1	The Collection Transaction Extension supports the creation, editing, and deleting of Collections through POST, PUT, PATCH, and DELETE requests.
Fields	STAC API - Features, STAC API - Item Search	Candidate	1.0.0	The Fields Extensions describes a mechanism to include or exclude certain fields from a response.
Filter	STAC API - Features, STAC API - Item Search	Candidate	1.0.0-rc.4	The Filter extension provides an expressive mechanism for searching based on Item attributes.
Transaction	STAC API - Features	Candidate	1.0.0-rc.3	The Transaction Extension supports the creation, editing, and deleting of items through POST, PUT, PATCH, and DELETE requests.

Pilot Extensions

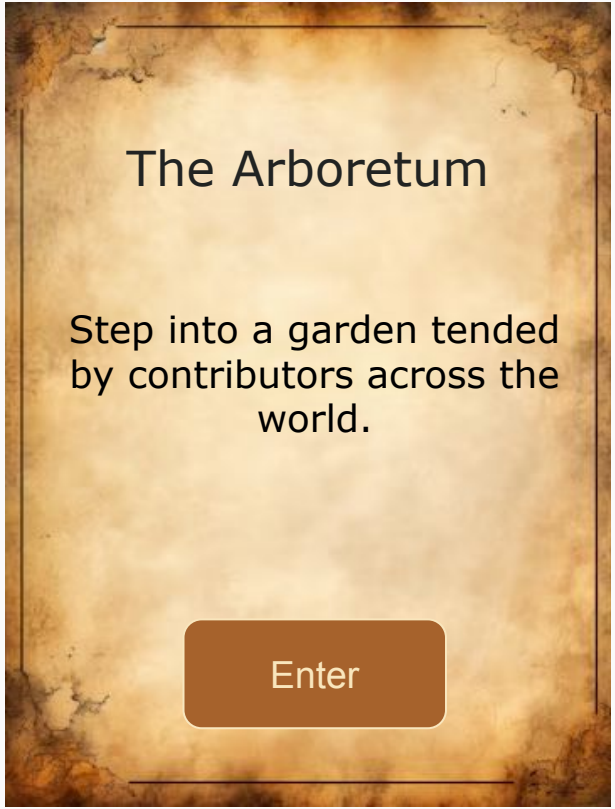
Collection Search	STAC API - Core	Pilot	1.0.0-rc.1	Collection Search for STAC APIs
Language (I18N)	STAC API - Core	Pilot	1.0.0-rc.2	Definitions and recommendations around making multi-lingual STAC APIs available

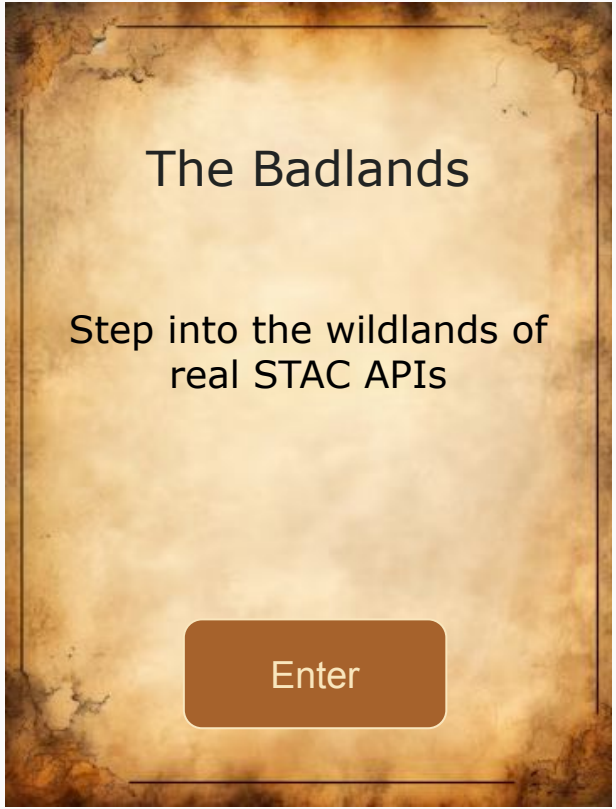
Proposed Extensions

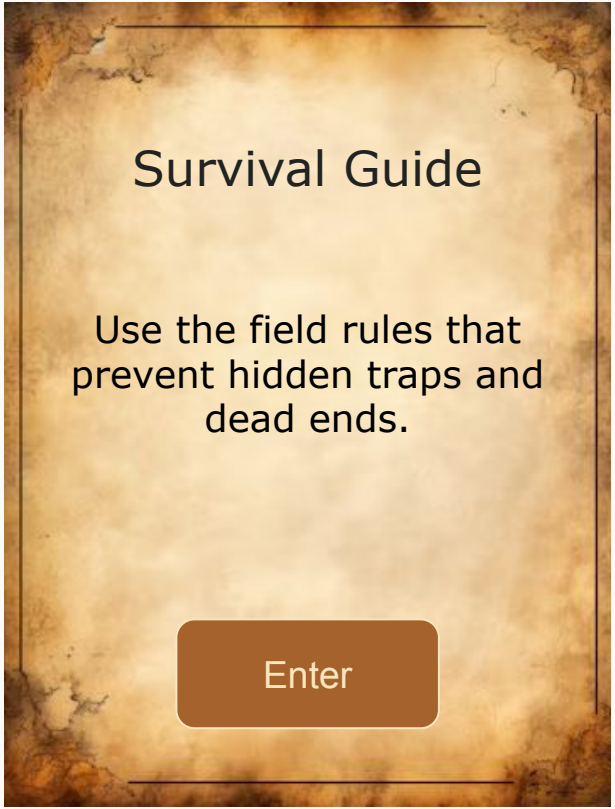
Aggregation	STAC API - Core	Proposal	0.3.0	Aggregation Extension to provide aggregated data over a search, rather than individual Item results
Browseable	STAC API - Core	Proposal	1.0.0-rc.3	Browseable advertises all Items in a STAC API Catalog can be reached by traversing child and item links.
Children	STAC API - Core	Proposal	1.0.0-rc.2	The purpose of this extension is to add an endpoint that presents a single resource from which clients can retrieve the immediate children of a Catalog.
Free-Text Search	- STAC API - Collection Search	Proposal	1.0.0-rc.1	This defines a new parameter, q that allows the user to perform free-text queries against the item properties.

API Extensions
(https://stac-api-extensions.github.io/)

- 14 Extensions in total
- Moved from API specification to stac-api-extensions GitHub organization
- New extensions:
  - Aggregation
  - Collection Search
  - Language
- Further alignment with OGC APIs (e.g. Filter/CQL2)

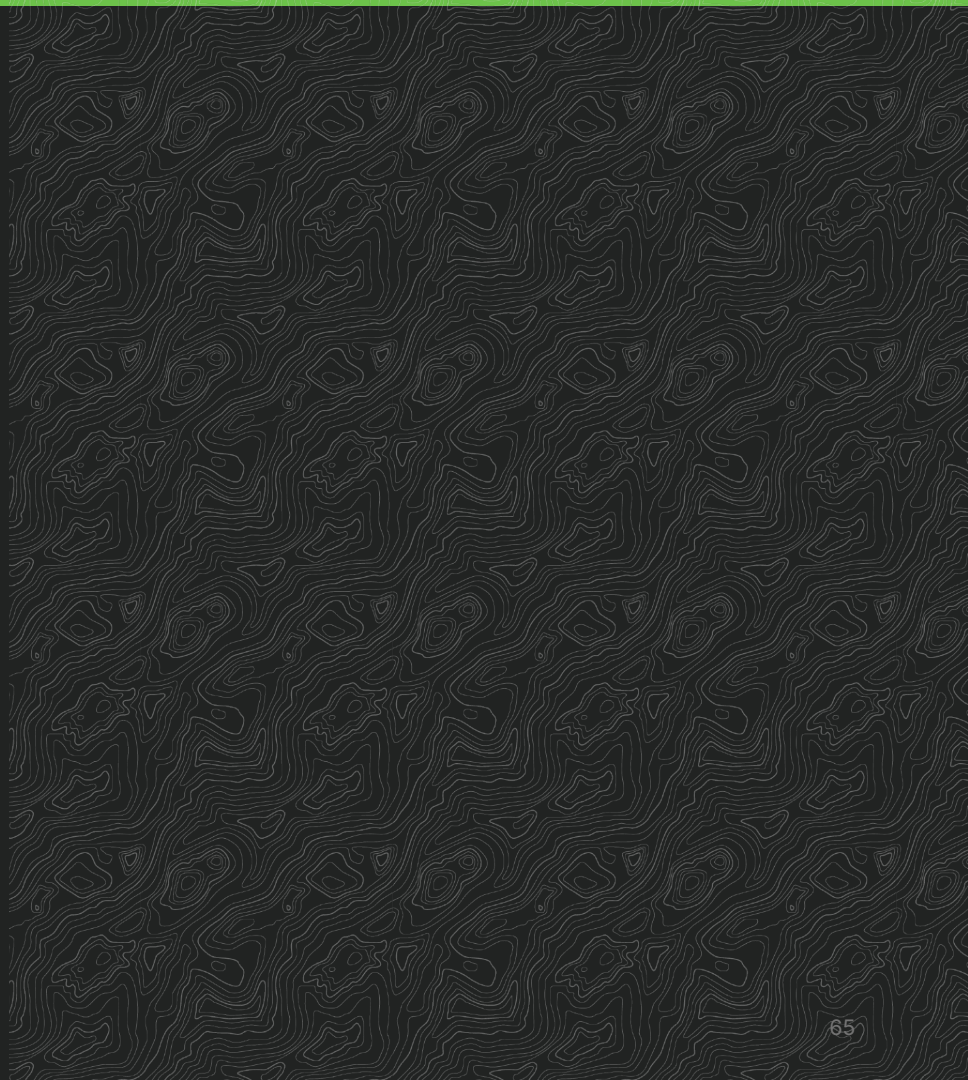






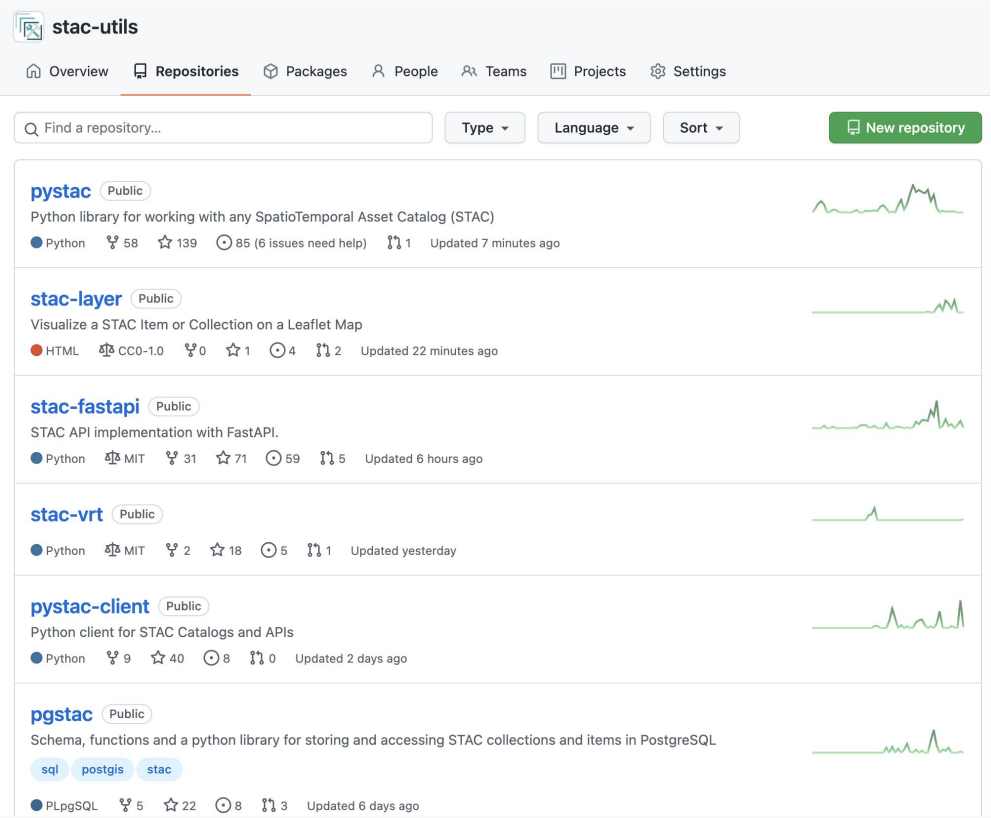
Explore the Cave

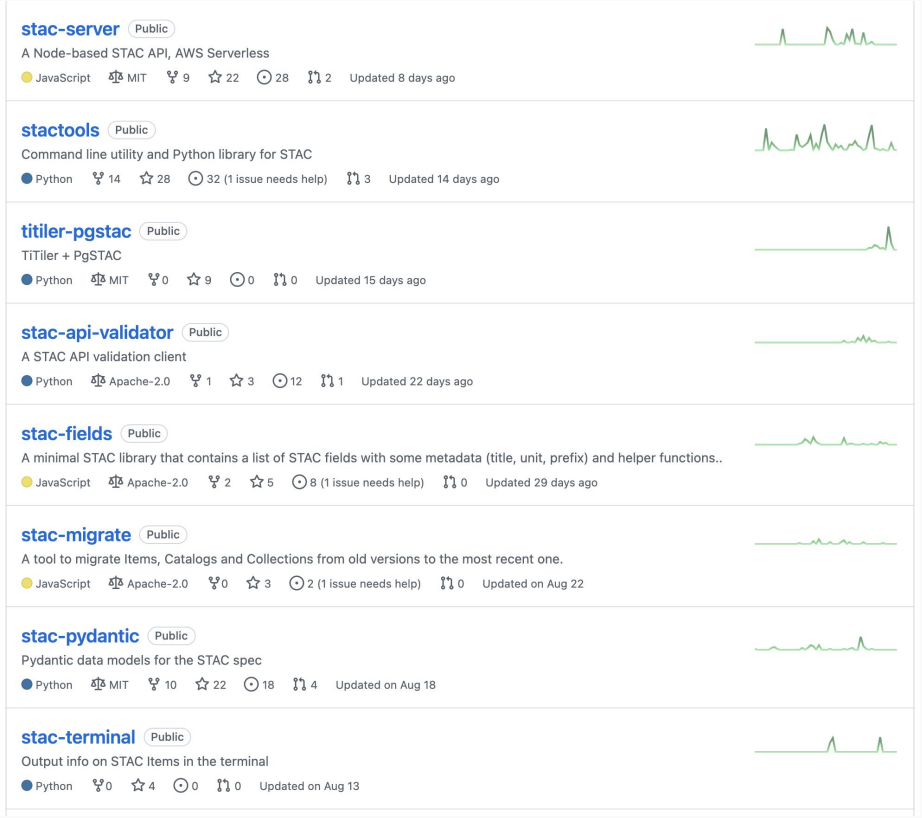
STAC Ecosystem



Element 84

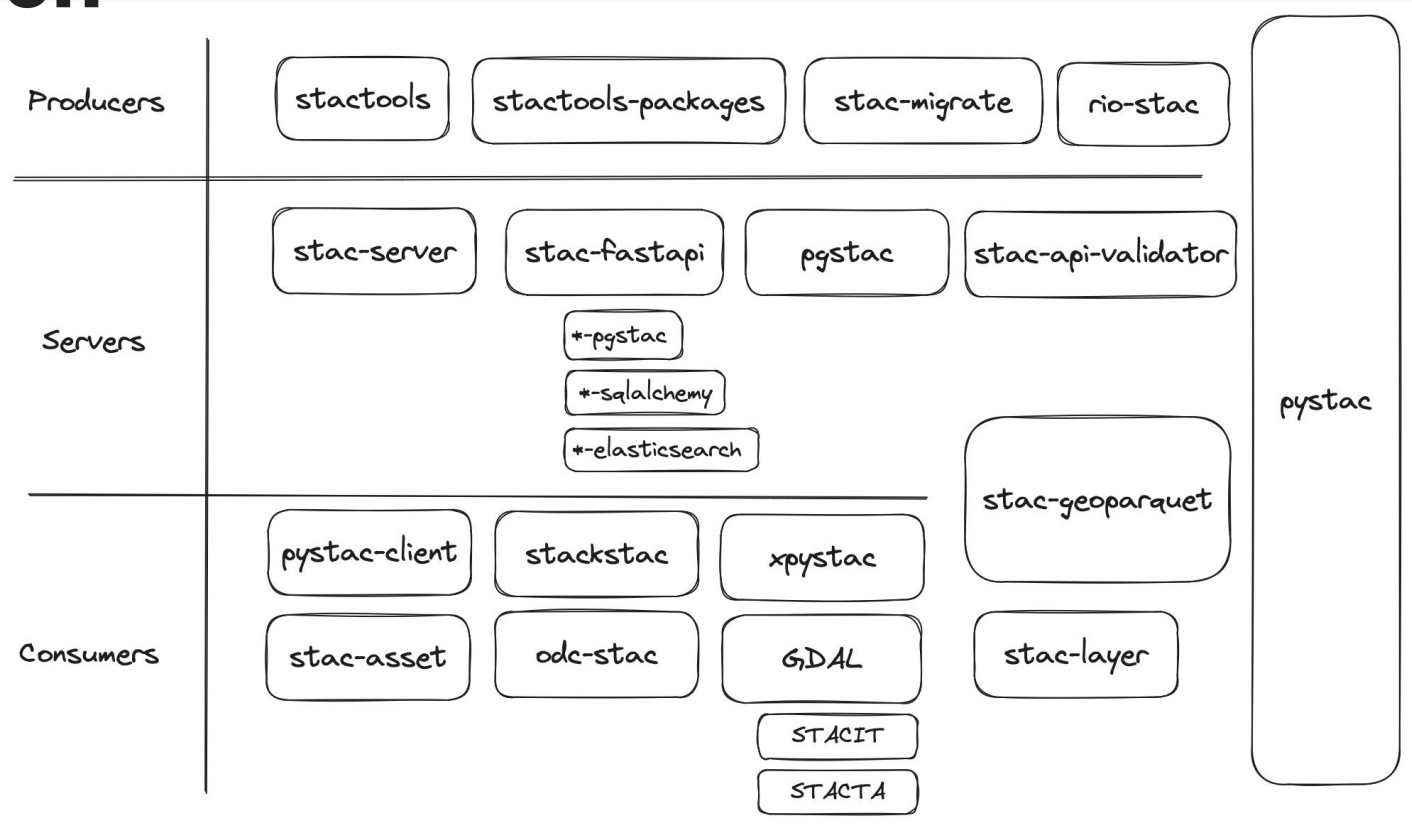
stac-utils





Element 84

Python


Producers
stactools
stactools-packages
stac-migrate
rio-stac
Servers
stac-server
stac-fastapi
pgstac
stac-api-validator
*-pgstac
*-sqlalchemy
*-elasticsearch
stac-geoparquet
pystac
Consumers
pystac-client
stackstac
xpystac
stac-asset
odc-stac
GDAL
stac-layer
STACIT
STACTA

pystac-client

A Python client for working with STAC APIs.

Installation

PySTAC Client is published to PyPi as pystac-client.

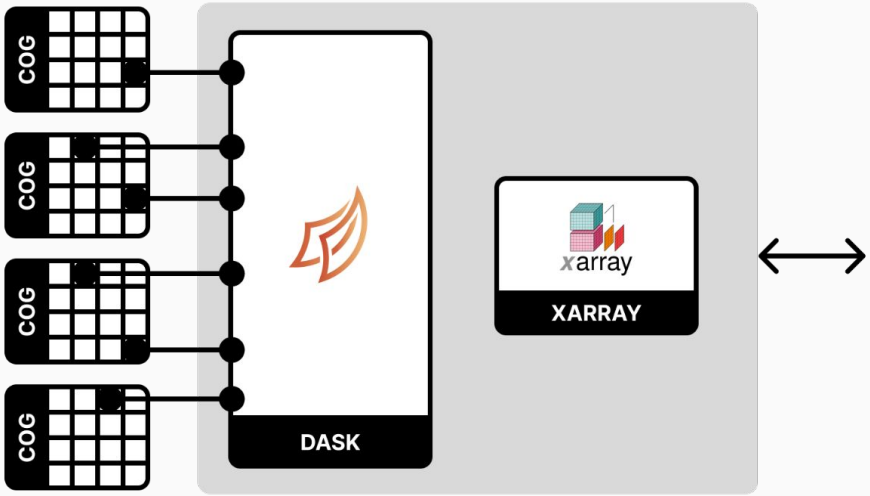
The only direct Python dependencies of **pystac-client** are PySTAC, requests, and dateutil.

To install with pip, run:

```
python -m pip install pystac-client
```

OpenDataCube + STAC

[odc-stac]



Find STAC Items to Load

```
[3]: km2deg = 1.0 / 111
     x, y = (113.887, -25.843)  # Center point of a query
     r = 100 * km2deg

     query = catalog.search(
         collections=["sentinel-s2-l2a-cogs"],
         datetime="2021-09-16",
         limit=10,
         bbox=(x - r, y - r, x + r, y + r),
     )

     items = list(query.get_items())
     print(f"Found: {len(items):d} datasets")
```

```
Found: 9 datasets
```

Construct Dask Dataset

Note that even though there are 9 STAC Items on input, there is only one timeslice on output. This is because of `groupy="solar_day"`. With that setting `stac_load` will place all items that occured on the same day (as adjusted for the timezone) into one image plane.

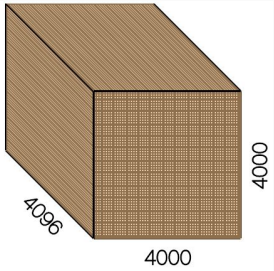
```
[4]: # crs = "epsg:32749"  # native UTM projection in the query region
     crs = "epsg:3857"  # Since we will plot it on a map we need to use 3857 projection
     zoom = 2 ** 5  # overview level 5

     xx = stac_load(
         items,
         output_crs=crs,
         resolution=(-10 * zoom, 10 * zoom),
         chunks={},
         groupby="solar_day",
         measurements=["red", "green", "blue"],
         stac_cfg=cfg,
     )
     xx
```

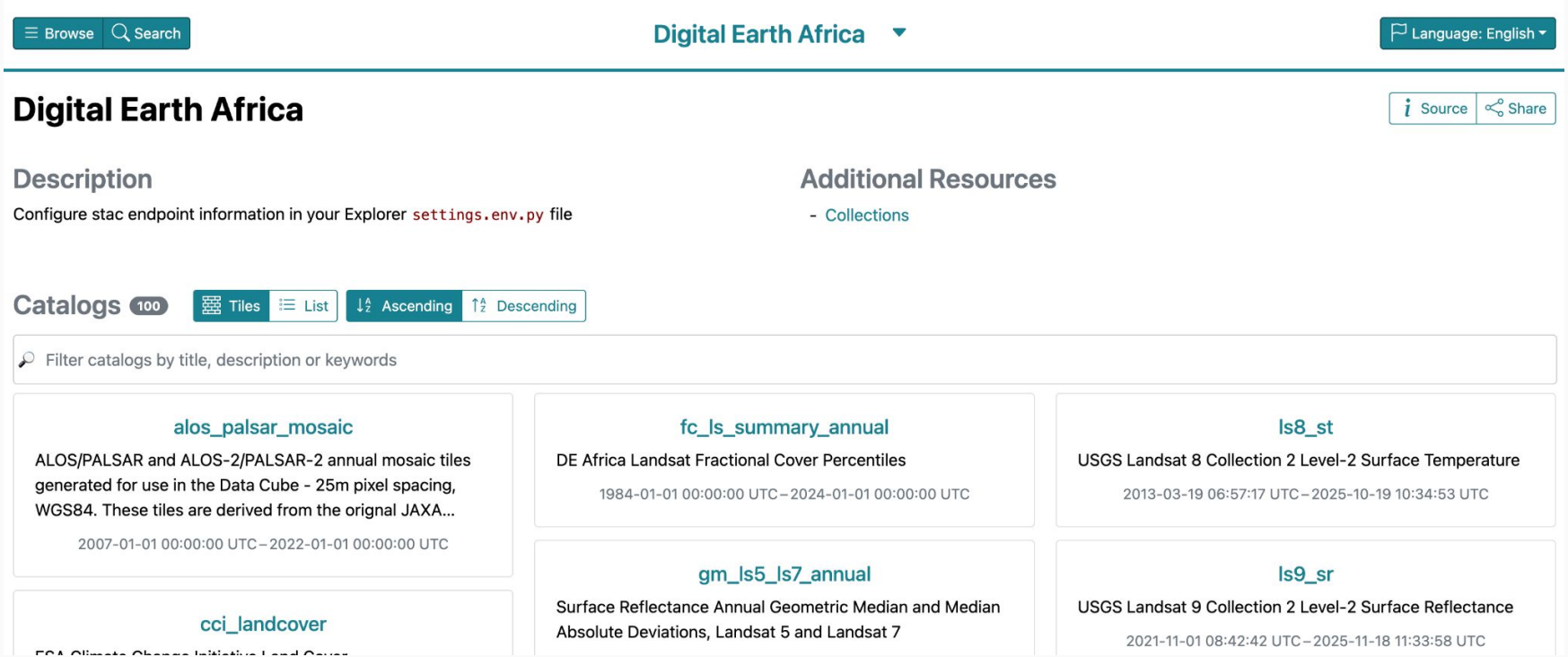
[4]: xarray.Dataset

▸ Dimensions: (time: 1, y: 1098, x: 833)

	Array	Chunk
Bytes	524.29 GB	125.96 MB
Shape	(4096, 4000, 4000)	(4096, 62, 62)
Count	258983 Tasks	4225 Chunks
Type	float64	numpy.ndarray



STAC Browser



https://radiantearth.github.io/stac-browser

JavaScript

- stac-js: lightweight drop-in classes for STAC objects
- stac-layer rewrite
- STAC implementation for OpenLayers (PR)
- STAC Browser v3
- stac-server
    - ready for STAC API 1.0.0
    - support for the Aggregation Extension and OpenSearch

What should I use to ‘serve’ STAC?

- None?
  - Static catalog with linked JSON
  - GeoParquet files (STAC geoparquet)
- Which API server should I use
  - stac-fastapi
  - stac-server
  - Pygeoapi
- Backend database?
  - PostGIS
  - OpenSearch/Elastic?

OpenSearch – Pros

👍 Advantage
Very fast for bbox and intersects searches
Scales easily to millions of items
Supports text search and metadata filters
Great for map UIs and faceted browsing

Con: Can be expensive to run at scale; intersections are approximate.

PostGIS – Pros

👍 Advantage
Exact geometry and topology (no approximations)
Supports complex spatial joins and analytics
Good for ingestion QA and coverage analysis
Lower operational cost for smaller workloads

Con: Slower and harder to scale for high-volume searches.

Blacksmith’s Anvil

Chain workflows to forge new outputs and capture their lineage.

Enter

The Vector Trail

Explore how STAC relates to vector layers, features, and shapes

Enter

Survival Guide

Use the field rules that prevent hidden traps and dead ends.

Read

Explore the Cave

STAC Best Practices

https://github.com/radiantearth/stac-spec/blob/master/best-practices.md

- **Web Best Practices**
  - Enable Cross-origin resource sharing (CORS)
  - STAC on the Web
    - Schema.org, JSON-LD, DCAT, microformats, etc
    - Deploying STAC Browser
  - Requester Pays
  - Consistent URIs
- **Item Best Practices**
  - Field and ID formatting
  - Searchable Identifiers
  - Field selection and Metadata Linking
  - Datetime selection
  - Unlocated Items
  - Representing Vector Layers in STAC
- **Asset and Link Best Practices**
  - Common Use Cases of Additional Fields for Assets
  - Working with Media Types
  - Asset Roles
  - Bands
- **Catalog & Collection Best Practices**
  - Static and Dynamic Catalogs
    - Static Catalogs
    - Dynamic Catalogs
  - Catalog Layout
    - Dynamic Catalog Layout
    - Mixing STAC Versions
  - Using Summaries in Collections
  - Use of links
    - Self-contained Catalogs
    - Published Catalogs
  - Using Relation Types
  - Versioning for Catalogs
  - Static to Dynamic best practices
    - Ingestion and links
    - Keep catalogs in sync with cloud notification and queue services
  - How to Differentiate STAC Files

STAC Best Practices

> ⚠ Warning
>
> This repository is a very early version and not officially endorsed by the STAC PSC yet. We plan to also move the best practices from the official stac-spec and stac-api-spec repositories over to this repository in the future.
>
> Feedback is always welcome, ideally via GitHub issues or pull requests.

This repository contains the STAC Best Practices.

This repository makes a number of recommendations for creating real world SpatioTemporal Asset Catalogs. None of them are required to meet the core specification, but following these practices will make life easier for client tooling and for users. They come about from practical experience of implementors and introduce a bit more 'constraint' for those who are creating STAC objects representing their data or creating tools to work with STAC.

While the current goal of the core is to remain quite flexible and simple to meet a wide variety of use cases, in time some of these may evolve to become part of the core specification.

STAC Specification

- Web Best Practices
- Item Best Practices
- Asset and Link Best Practices
- Catalog and Collection Best Practices
- Metadata and Extension Best Practices

https://github.com/radiantearth/stac-best-practices/

Collection Fields - 1

id	Include data type and level (i.e., product) in the id. Use lowercase and dashes for separators.
stac_version	1.1.0
title	Human readable title
description	Detailed description of the data collection, use CommonMark
keywords	Such as sensor modality (e.g., sar, optical), type of data (e.g., NITF, COG)
stac_extensions	This may be an empty array, unless summaries are included in which case the Extensions are any extensions an Item uses whose fields are included in the summaries dictionary.
license	An SPDX License identifier if there is one, or `other`. If other then you must include a link with a relational type of `license` that points to the license file. This should be included alongside the `collection.json` file.
providers	Include all providers and processors of the data.

Collection Fields - 2

extent	It is ok to just make the spatial extension the whole world if the extent is not known. The temporal extent may not be known and could change as Items are added so it can just be an approximation. The extents can be updated later if needed.
summaries	Summaries can be useful in providing a detailed description of the data within the Collection, however may not initially be known so can be left out.
links	Include any internal (e.g., Confluence page) or external link (such as a vendor product page or docs) that provides additional context.
assets	These are data assets for the Collection, it is not any per item assets. This may be something like a representative image to be used as a thumbnail in a UI. Collection Assets are not required.
item_assets	One of the most important fields to include this should contain all of the static asset entry fields that can be found within items in the Collection. The `item_assets` dictionary will look the same as any Item `assets` dictionary except the Collection will not have URLs to a data asset or any fields that are specific to an Item.

What extensions should I use in Items?

Extension	Minimum fields to include in Item properties
eo	eo:cloud_cover, eo:snow_cover
grid	grid:code
processing	processing:level, processing:datetime, processing:software at a minimum
product	product:type only if using the sar extension
projection	proj:code (if rectified data), proj:centroid
storage	storage:schemes to define type, platform, and region
view	view:off_nadir, view:azimuth and whatever other angles are available
sar	Include any and all fields that are available

Asset-level Extensions

The following extension fields are defined at the Asset level rather than the Item level. If all assets within an Item share the exact same value then it could be defined in Item properties, but this is rare for these fields.

Extension	Minimum fields to include in Asset objects
eo	eo:common_name, eo:center_wavelength, eo:full_width_half_max
file	file:checksum, file:size
projection	proj:shape, proj:transform
raster	raster:scale, raster:offset
storage	Storage:refs to reference storage:schemes in Item properties

STAC Profiles from CEOS-ARD

STAC Extensions

This profile doesn't define new STAC fields, it's just a profile that uses existing STAC extensions to map and fulfill the CEOS-ARD requirements.

As the identifier for this profile is just a collection of existing extensions and only defines required fields, you get a good bit of validation already without providing the identifier/schema of this profile in `stac_extensions`. If your metadata is already compliant to this profile, you can omit the identifier for this profile in `stac_extensions` to avoid costly regeneration of the Items. You won't get validation whether all required fields are present, but this could be checked manually in the CEOS assessment/review.

The following STAC extensions are relevant for this profile:

Name	Schema URI for `stac_extensions`	Required
Accuracy	`https://stac-extensions.github.io/accuracy/v1.0.0-beta.1/schema.json`	✕
CEOS-ARD	`https://stac-extensions.github.io/ceos-ard/v0.2.0/optical/schema.json`	✓
Classification	`https://stac-extensions.github.io/classification/v2.0.0/schema.json`	✕
Electro Optical	`https://stac-extensions.github.io/eo/v2.0.0/schema.json`	✓
Processing	`https://stac-extensions.github.io/processing/v1.2.0/schema.json`	✕
Projection	`https://stac-extensions.github.io/projection/v2.0.0/schema.json`	✓
Raster	`https://stac-extensions.github.io/raster/v2.0.0/schema.json`	✓
View Geometry	`https://stac-extensions.github.io/view/v1.1.0/schema.json`	✓

Blacksmith’s Anvil

Chain workflows to forge new outputs and capture their lineage.

Enter

The Badlands

Step into the wildlands of real STAC APIs

Enter

The Tower

Ascend the tower to extend your catalog with content and features

Ascend

Explore the Cave

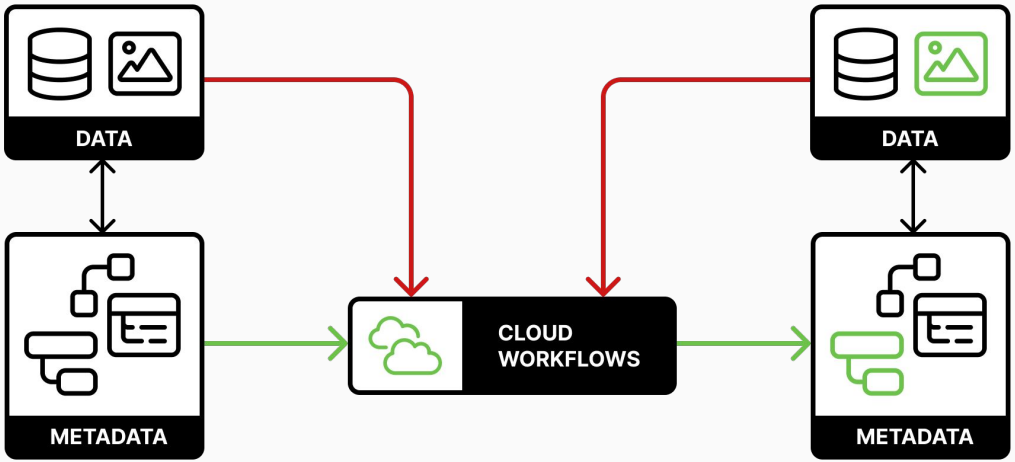
STAC Workflows

Stop thinking of working on files

```
LC08_L1GT_168005_20190828_20190828_01_RT.tif
LC08_L1GT_168005_20190828_20190828_01_RT_B1.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_B3.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_c10_f1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_cm.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_class_c50_i9.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_tmp_test5.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_ver7.6.tif
LC08_L1GT_168005_20190828_20190828_01_final_class2.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_class.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_final.tif
LC08_L1GT_168005_20190828_20190828_01_test.tif
LC08_L1GT_168005_20190828_20190828_test1.tif
LC08_L1GT_168005_20190828_20190828_test_test.tif
```

Cloud-Native Geospatial Workflows

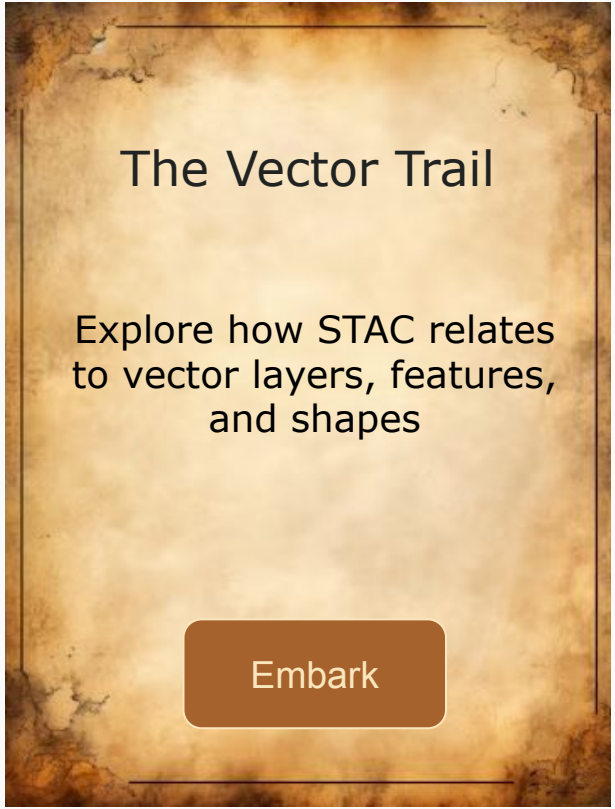
- Workflows operate on metadata
  - Metadata in, metadata out (STAC In, STAC out)
- Metadata contains complete characterization of data
  - Data location and access

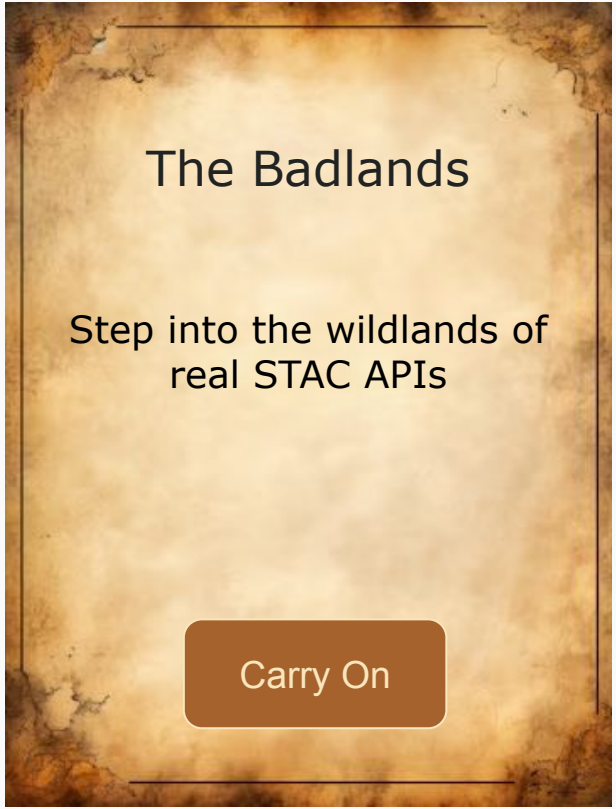


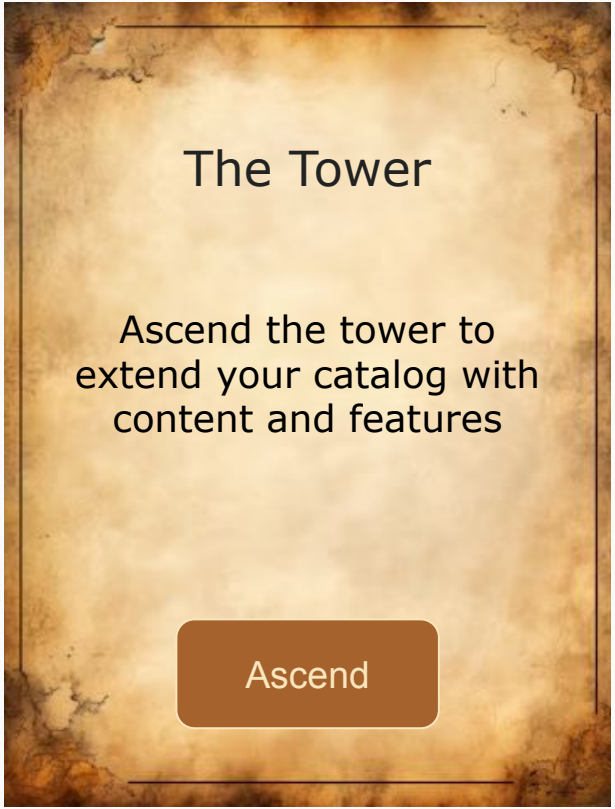
Data provenance


```
"properties": {
  "datetime": "2016-08-22T18:28:23.368922Z",
  "start_datetime": "2016-08-22T18:28:23.368922Z",
  "end_datetime": "2016-08-22T18:28:48.368201Z",
  "created": "2016-08-23T00:38:22Z",
  "platform": "sentinel-1a",
  "constellation": "sentinel-1",
  "mission": "sentinel-1",
  "instruments": [
    "c-sar"
  ],
  "sat:absolute_orbit": 12717,
  "sat:orbit_state": "ascending",
  "sat:relative_orbit": 45,
  "sat:anx_datetime": "2016-08-22T18:24:52.513706Z",
  "sar:instrument_mode": "IW",
  "sar:frequency_band": "C",
  "sar:polarizations": [
    "VV",
    "VH"
  ],
  "sar:product_type": "GRD",
  "processing:lineage": "GRD Post Processing",
  "processing:level": "L1C",
  "processing:facility": "Copernicus S1 Core Ground Segment - DPA",
  "processing:software": {
    "Sentinel-1 IPF": "002.71"
  }
```

```
    "links": [
      {↔},
      {↔},
      {↔},
      {
        "title": "Source STAC Item",
        "rel": "derived_from",
        "href": "https://cirrus-v0-data-1qm7gekzjucbq.s3.us-west-2.amazonaws.com/sentinel-s2-l2a/53/Q/PB/2022/7/S2A_53QPB_20220721_0_L2A/S2A_53QPB_20220721_0_L2A.json",
        "type": "application/json"
      },
      {↔},
      {
        "rel": "parent",
        "href": "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
      },
      {
        "rel": "collection",
        "href": "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
      },
      {
        "rel": "root",
        "href": "https://earth-search.aws.element84.com/v0/"
      }
    ]
  },
```



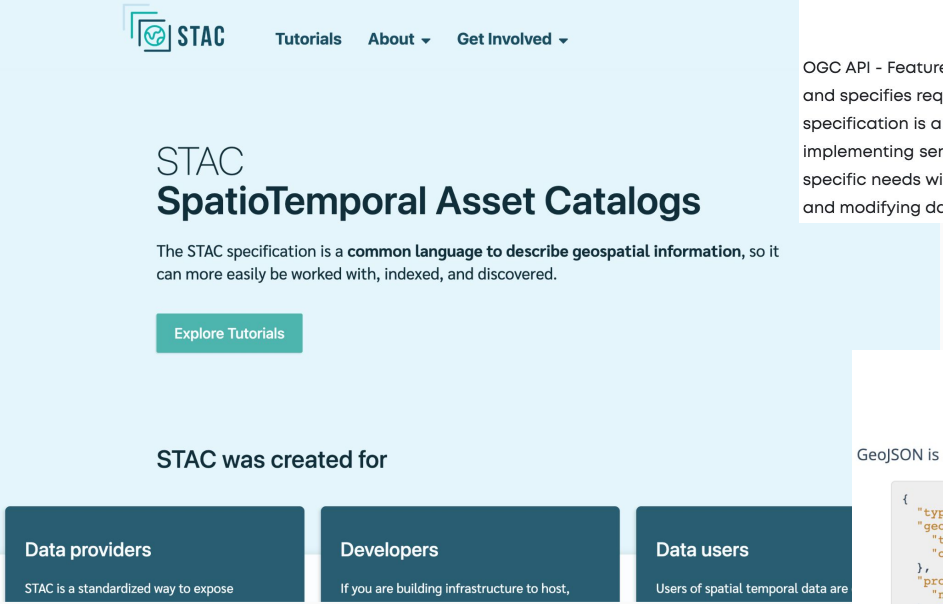




Explore the Cave

STAC for vector data

STAC vs OGC API - Features



OGC API - FEATURES

OGC API - Features is a multi-part standard that offers the capability to create, modify, and query spatial data on the Web and specifies requirements and recommendations for APIs that want to follow a standard way of sharing feature data. The specification is a multi-part document. The Core part of the specification describes the mandatory capabilities that every implementing service has to support and is restricted to read-access to spatial data. Additional capabilities that address specific needs will be specified in additional parts. Envisaged future capabilities include, for example, support for creating and modifying data, more complex data models, richer queries, and additional coordinate reference systems.

GeoJSON

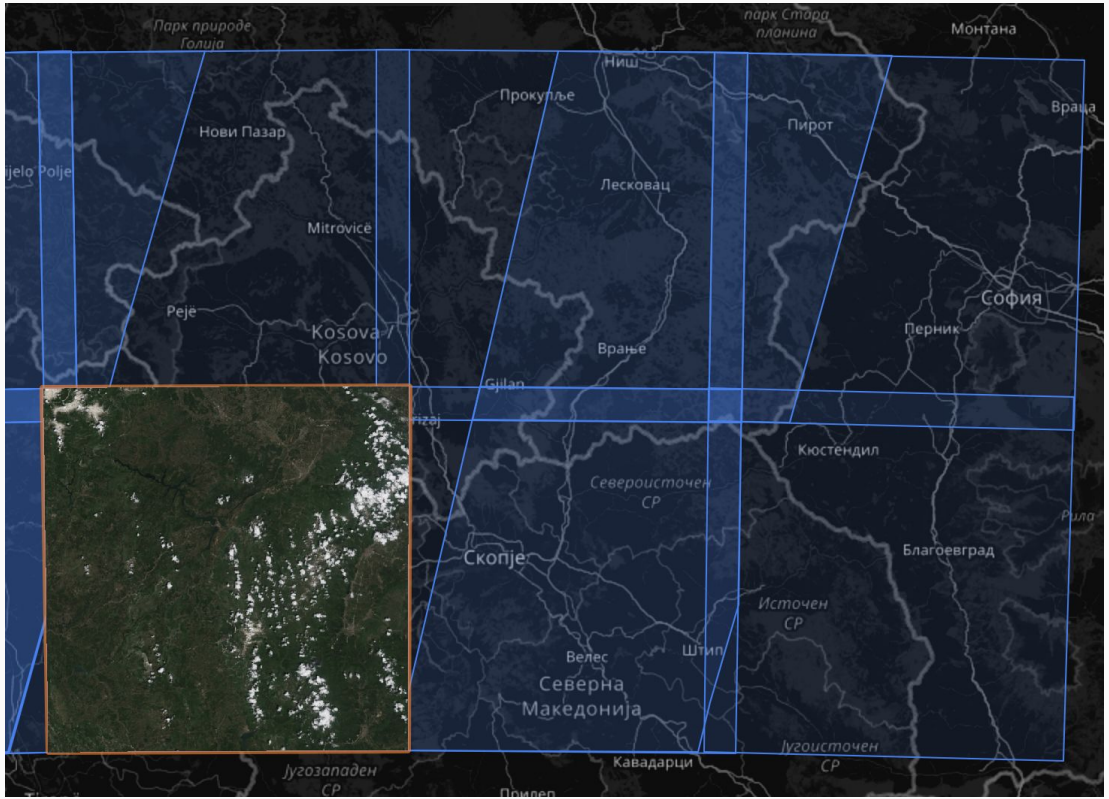
GeoJSON is a format for encoding a variety of geographic data structures.

```
{
  "type": "Feature",
  "geometry": {
    "type": "Point",
    "coordinates": [125.6, 10.1]
  },
  "properties": {
    "name": "Dinagat Islands"
  }
}
```

GeoJSON supports the following geometry types: `Point`, `LineString`, `Polygon`, `MultiPoint`, `MultiLineString`, and `MultiPolygon`. Geometric objects with additional properties are `Feature` objects. Sets of features are contained by `FeatureCollection` objects.

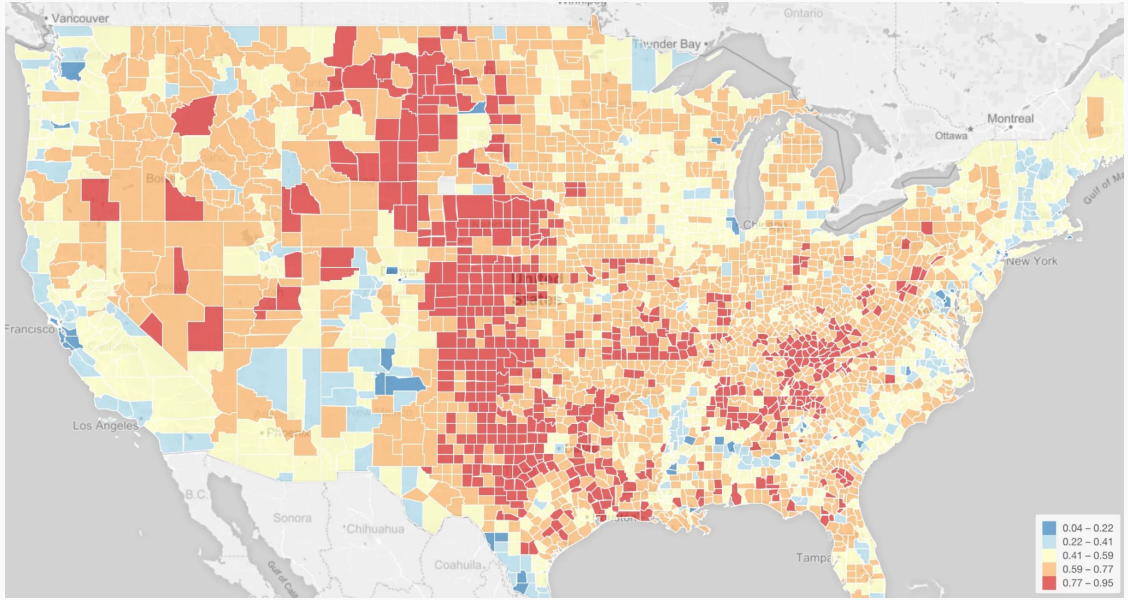
STAC Items

- GeoJSON
- Defines region where there is data
- May contain additional properties
- Data is in separate files (assets)



OGC Features

- GeoJSON
- Defines regions of uniform properties



Use OGC Features if geometry is the data

Use STAC if geometry defines where data is

Google-Microsoft Open Buildings - combined by VIDA

Overview

This dataset merges Google's V3 Open Buildings and Microsoft's latest Building Footprints. With 2,534,595,270 footprints, it is to our knowledge the most complete openly available dataset as of Sep. 2023. It covers 92% of Level 0 administrative boundaries, and is divided into 182 partitions. Each footprint is labelled with its respective source, either Google or Microsoft. It can be accessed in cloud-native geospatial formats such as GeoParquet, FlatGeobuf and PMTiles.

See it in action

You can Observable to get a quick overview of the dataset or go to VIDA to see it in action.

Original datasets

Google

The original Google V3 open buildings is downloadable from this link as gzipped CSV files. Here are some key details about the original dataset:

> *The dataset contains 1.8 billion building detections, across an inference area of 58M km2 within Africa, South Asia, South-East Asia, Latin America and the Caribbean.*
>
> *Each building in the dataset has a polygon defining its footprint on the ground, a confidence score indicating how certain we are that this is a building, and a Plus Code corresponding to the centre of the building. There is no information about the type of building, its street address, or any details other than its geometry.*

For more comprehensive information, please visit the description page. You can also check out the FAQ section for additional information.

Microsoft

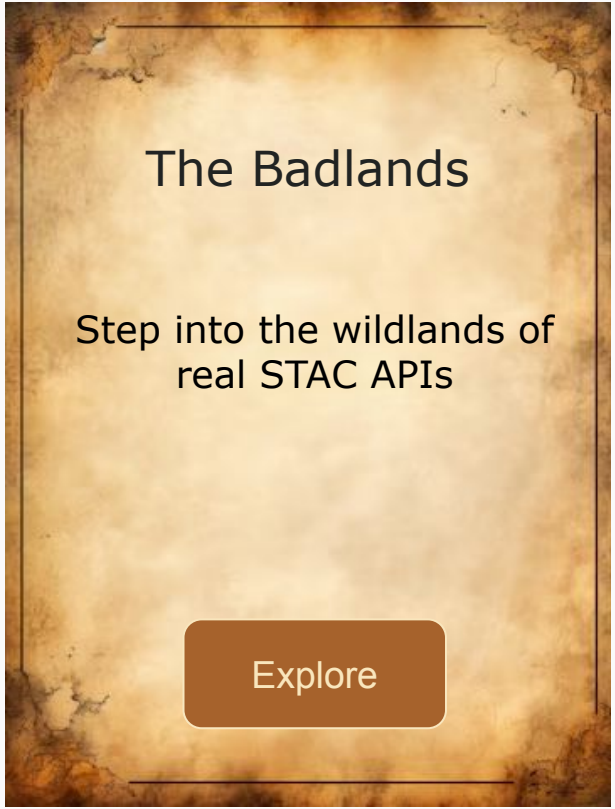
The latest version of Microsoft's building footprints can be downloaded from Microsoft Planetary Computer as gzipped partitioned files.

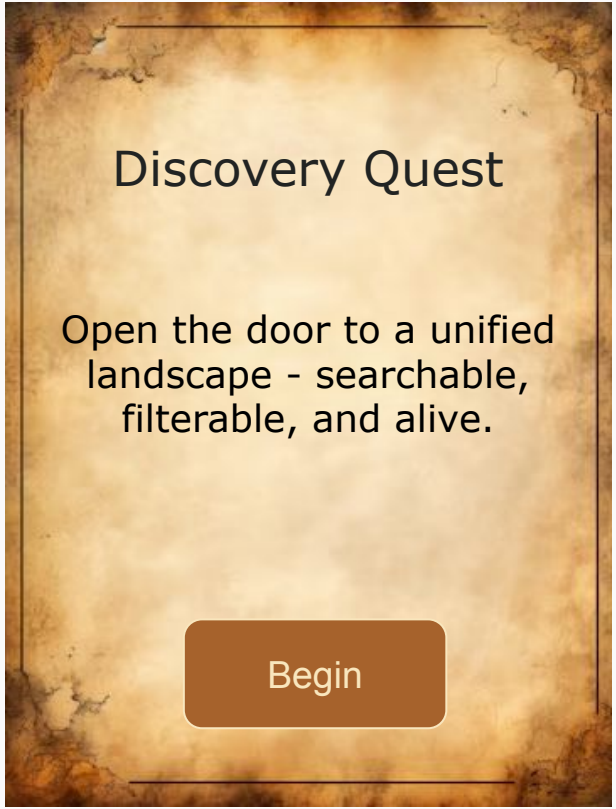
> *The Microsoft Global Open Buildings dataset was generated through Bing Maps, which detected a total of 1.24 billion buildings. These buildings were identified using imagery from Bing Maps, encompassing data collected between 2014 and 2023, including images from Maxar, Airbus, and IGN France.*

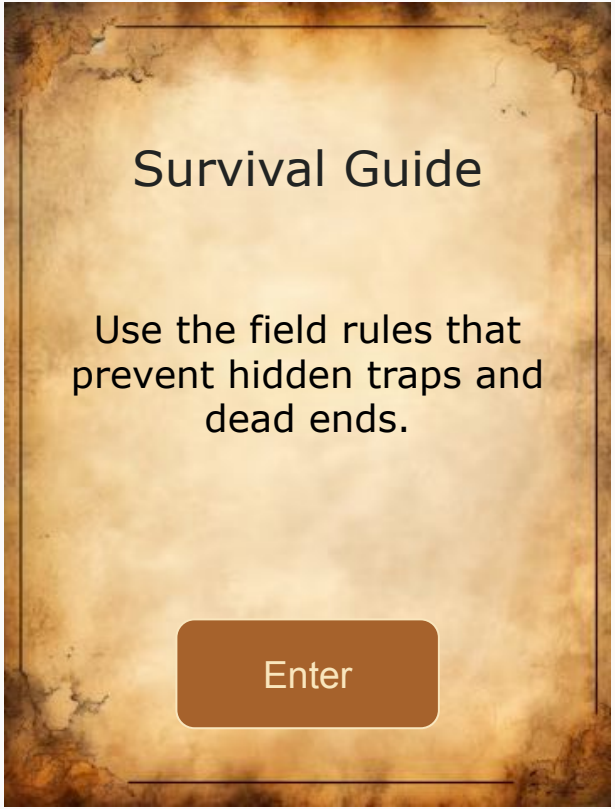
For more detailed information, please visit the github page.

STAC for partitioned vector data

- Very large datasets
  - Avoid expense and complexity of scaling Features API
- Data provider/publisher
  - Focus on creating and distribution, not usage
- Archival
  - Publishing dataset for reproducibility (e.g., scientific study)
- Data Formats
  - FlatGeoBuf
  - GeoParquet
  - PMTiles





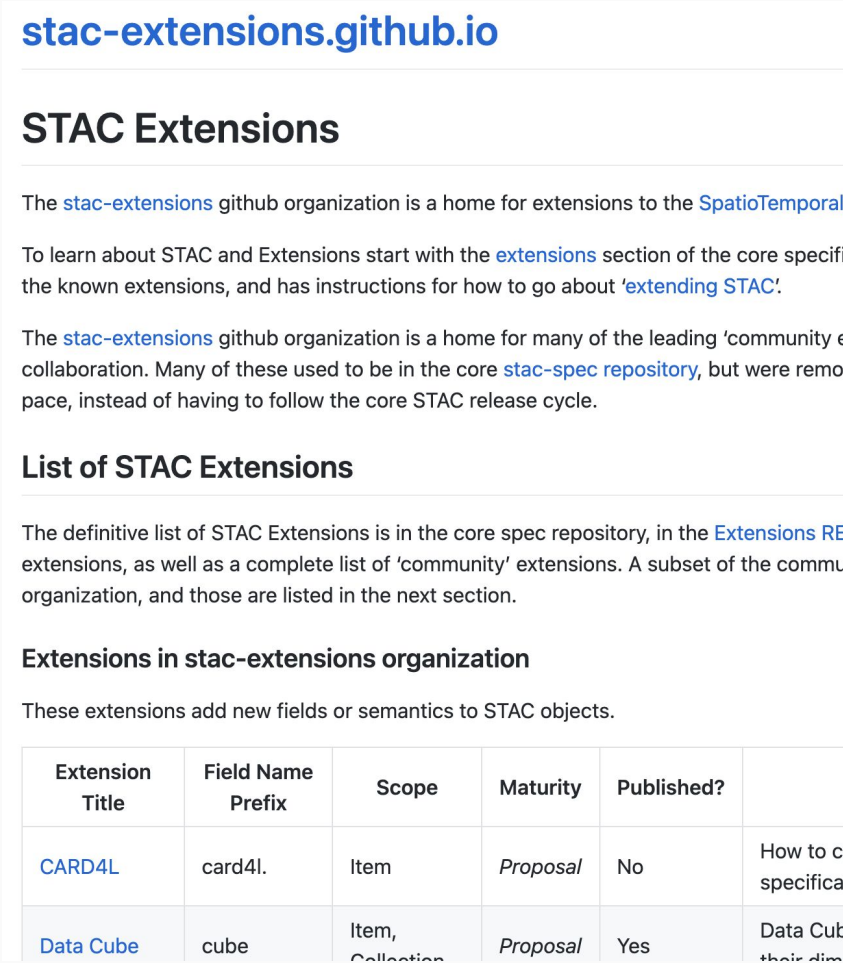


Explore the Cave

The Extension landscape

Extensions

- 78 Extensions in the definitive list
  - https://stac-extensions.github.io/
- 66 hosted in GitHub organization
  - https://github.com/stac-extensions
- Extensions rated by “Maturity”
  - Stable
  - Candidate
  - Pilot
  - Proposal
  - Deprecated



stac-extensions.github.io

STAC Extensions

The stac-extensions github organization is a home for extensions to the SpatioTemporal

To learn about STAC and Extensions start with the extensions section of the core specifi
the known extensions, and has instructions for how to go about ‘extending STAC’.

The stac-extensions github organization is a home for many of the leading ‘community e
collaboration. Many of these used to be in the core stac-spec repository, but were remo
pace, instead of having to follow the core STAC release cycle.

List of STAC Extensions

The definitive list of STAC Extensions is in the core spec repository, in the Extensions RE
extensions, as well as a complete list of ‘community’ extensions. A subset of the commu
organization, and those are listed in the next section.

Extensions in stac-extensions organization

These extensions add new fields or semantics to STAC objects.

Extension Title	Field Name Prefix	Scope	Maturity	Published?	
CARD4L	card4l.	Item	*Proposal*	No	How to c specifica
Data Cube	cube	Item, Collection	*Proposal*	Yes	Data Cub their dim

Extensions by Maturity

Stable

- Electro-Optical
- File Info
- Landsat
- Projection
- Scientific Citation
- View Geometry

Candidate

- Datacube
- Processing
- Raster
- SAR
- Satellite
- Sentinel-2
- Versioning Indicators

Deprecated

- Deep Learning Model*
- Item Assets Definition
- Sentinel-3
- Single File STAC
- Time Series

Pilot Extensions

- Alternate Assets
- Anonymized Location
- CARD4L (Optical and SAR)
- Classification
- Grid
- Label
- Machine Learning Model Extension*
- Military Grid Reference System
- NOAA Geostationary Operational Environmental Satellite (GOES)
- NOAA MRMS QPE
- Order
- Point Cloud
- Stats
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- Timestamps
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Proposal Extensions

- Accuracy
- Altimetry
- Attribution
- Authentication
- CEOS-ARD
- CF
- Contacts
- Disasters Charter*
- Example Links
- Forecast
- InSAR
- Language (I18N)
- ML AOI
- ML Model
- Open Science Catalog
- Perspective Imagery
- Product
- Quality*
- Rendering
- Sentinel-1
- Sentinel-5P
- Solar System
- Stereo Imagery
- Themes
- Tiled Assets
- TrainingDML-AI*
- U.S. Fish & Wildlife Service (FWS) National Wetlands Inventory (NWI)
- Video
- Virtual Assets
- Web Map Links

WIP Extensions

- Aerial Photography*
- Archive
- CMIP6
- Camera*
- Composite
- Earthquake
- Film*
- Hyperspectral Imagery
- Scanning*

“Content” Extensions

Electro-Optical

Field Name	Type	Description
eo:bands	[Band Object]	An array of available bands where each object is a Band Object. If given, requires at least one band.
eo:cloud_cover	number	Estimate of cloud cover

SAR

Field Name	Type	Description
sar:instrument_mode	string	**REQUIRED.** The name of the sensor acquisition mode that is commonly used. This should be the short name, if available. For example, `WV` for "Wave mode" of Sentinel-1 and Envisat ASAR satellites.
sar:frequency_band	string	**REQUIRED.** The common name for the frequency band to make it easier to search for bands across instruments. See section "Common Frequency Band Names" for a list of accepted names.
sar:center_frequency	number	The center frequency of the instrument, in gigahertz (GHz).
sar:polarizations	[string]	**REQUIRED.** Any combination of polarizations.
sar:product_type	string	**REQUIRED.** The product type, for example `SSC`, `MGD`, or `SGC`
sar:resolution_range	number	The range resolution, which is the maximum ability to distinguish two adjacent targets perpendicular to the flight path, in meters (m).
sar:resolution_azimuth	number	The azimuth resolution, which is the maximum ability to distinguish two adjacent targets parallel to the flight path, in meters (m).
sar:pixel_spacing_range	number	The range pixel spacing, which is the distance between adjacent pixels perpendicular to the flight path, in meters (m). Strongly RECOMMENDED to be specified for products of type `GRD`.
sar:pixel_spacing_azimuth	number	The azimuth pixel spacing, which is the distance between adjacent pixels parallel to the flight path, in meters (m). Strongly

View

Field Name	Type	Description
view:off_nadir	number	The angle from the sensor between nadir (straight down) and the scene center. Measured in degrees (0-90).
view:incidence_angle	number	The incidence angle is the angle between the vertical (normal) to the intercepting surface and the line of sight back to the satellite at the scene center. Measured in degrees (0-90).
view:azimuth	number	Viewing azimuth angle. The angle measured from the sub-satellite point (point on the ground below the platform) between the scene center and true north. Measured clockwise from north in degrees (0-360).
view:sun_azimuth	number	Sun azimuth angle. From the scene center point on the ground, this is the angle between truth north and the sun. Measured clockwise in degrees (0-360).
view:sun_elevation	number	Sun elevation angle. The angle from the tangent of the scene center point to the sun. Measured from the horizon in degrees (`-90` - `90`). Negative values indicate the sun is below the horizon, e.g. sun elevation of -10° means the data was captured during nautical twilight.

“Usage” Extensions

projection

Field Name	Type	Description
proj:epsg	integer|null	**REQUIRED.** EPSG code of the datasource
proj:wkt2	string|null	WKT2 string representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:projjson	PROJJSON Object|null	PROJJSON object representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:geometry	GeoJSON Geometry Object	Defines the footprint of this Item.
proj:bbox	[number]	Bounding box of the Item in the asset CRS in 2 or 3 dimensions.
proj:centroid	Centroid Object	Coordinates representing the centroid of the Item (in lat/long)
proj:shape	[integer]	Number of pixels in Y and X directions for the default grid
proj:transform	[number]	The affine transformation coefficients for the default grid

Raster (v2)

Field Name	Type	Description
raster:nodata	number|string	Pixel values used to identify pixels that are nodata in the band either by the pixel value as a number or `nan`, `inf` or `-inf` (all strings).
raster:sampling	string	One of `area` or `point`. Indicates whether a pixel value should be assumed to represent a sampling over the region of the pixel or a point sample at the center of the pixel.
raster:data_type	string	The data type of the pixels in the band. One of the data types as described below.
raster:bits_per_sample	number	The actual number of bits used for this band. Normally only present when the number of bits is non-standard for the `datatype`, such as when a 1 bit TIFF is represented as byte.
raster:spatial_resolution	number	Average spatial resolution (in meters) of the pixels in the band.
raster:statistics	Statistics Object	Statistics of all the pixels in the band.
raster:unit	string	Unit denomination of the pixel value.
raster:scale	number	Multiplicator factor of the pixel value to transform into the value (i.e. translate digital number to reflectance).
raster:offset	number	Number to be added to the pixel value (after scaling) to transform into the value (i.e. translate digital number to reflectance).
raster:histogram	Histogram Object	Histogram distribution information of the pixels values in the band.

storage

Field Name	Type	Description
storage:platform	string	The cloud provider where data is stored
storage:region	string	The region where the data is stored. Relevant to speed of access and inter region egress costs (as defined by PaaS provider)
storage:requester_pays	boolean	Is the data requester pays or is it data manager/cloud provider pays. *Defaults to false*
storage:tier	string	The title for the tier type (as defined by PaaS provider)

The Vector Trail

Explore how STAC relates to vector layers, features, and shapes

Embark

The Badlands

Step into the wildlands of real STAC APIs

Carry On

Survival Guide

Use the field rules that prevent hidden traps and dead ends.

Read

Explore the Cave

Exploring STAC Catalogs

Let’s dive into some STACs!

https://earth-search.aws.element84.com/v1
https://planetarycomputer.microsoft.com/api/stac/v1
https://cmr.earthdata.nasa.gov/stac
https://landsatlook.usgs.gov/stac-server

The Arboretum

Step into a garden tended by contributors across the world.

Enter

Discovery Quest

Open the door to a unified landscape - searchable, filterable, and alive.

Begin

Survival Guide

Use the field rules that prevent hidden traps and dead ends.

Enter

Explore the Cave

Challenges

STAC Challenges

- Sheer scale of metadata
- Flexibility is the enemy of interoperability
  - Best practices are often not practiced
  - Proliferation of extensions
- Tendency to shoehorn problems (and metadata) into STAC
  - OGC Features API vs STAC
- Interoperability across providers
  - Differences in metadata
  - Multi-cloud: Authoritative Sources vs Copies
- STAC Governance via steering committee

STAC Community

Cloud-Native Geospatial Forum (CNG)
6,386 followers
3w •

✨ Introducing the new **STACspec** Project Steering Committee (PSC) ✨ We've got some fresh faces joining the PSC, and we couldn't be more excited! Say hello to:

- **Alex Leith GAICD** (**Auspatious**)
- **Emmanuel Mathot** (**Development Seed**)
- **Julia Signell** (**Element 84**)
- **Michelle Roby** (**Radiant Earth**)
- **Peter Gadomski** (**Development Seed**)

They'll be joining **Matthias Mohr** (Independent) and **Matthew Hanson** (**Element 84**) in steering STAC's future.

🙏 We'd like to give a huge thank you to **Chris Holmes** (**Planet**), **James B.** (**Sparkgeo - Geospatial Consulting & Development**), and **Rob Emanuele** (**Microsoft**) for everything they've done to make STAC what it is today. You've helped build something incredible, and we're so grateful!

You have been eaten by a grue

Join the community meetings!

https://groups.google.com/a/cloudnativegeo.org/g/stac-community

Thank you!

Any more questions?

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Element 84